The Value of Vascular Information: Arterial Stiffness and Decision-Making in Hypertension Management

Abstract. Blood pressure (BP) is the primary measure used to manage hypertension and is widely observed in clinical practice. However, BP alone does not fully capture the condition of the arteries. Arterial stiffness (AS) provides additional insight by reflecting long-term structural changes in a patient's vascular system. In collaboration with the McGill University Health Centre (MUHC), this study examines whether arterial stiffness provides additional value in hypertension treatment decisions. To address our research question, we develop a Markov decision process (MDP) model for hypertension treatment that incorporates arterial stiffness as an additional physiological state variable. In our model, blood pressure and arterial stiffness jointly evolve over time, and arterial stiffness affects both cardiovascular risk and the effectiveness of antihypertensive treatment. We evaluate treatment policies through cohort simulations and compare the expected quality-adjusted life years (QALYs) with those generated by guideline-based policies.

Key words: hypertension management; arterial stiffness; pulse wave velocity; Markov decision process; dynamic optimization; risk-adjusted therapy; treatment policy; quality-adjusted life years

1. Introduction

Elevated blood pressure, hypertension, is a significant factor of global risk for premature death, heart attacks, and other cardiovascular diseases. According to the World Health Organisation (WHO), about 1.4 billion adults around the globe are living with hypertension. Of these adults, only a fraction has adequate treatment for elevated blood pressure, and approximately 20 to 25 percent have controlled blood pressure World Health Organization (2025). Inadequately controlled hypertension significantly raises the risk of major cardiovascular events, including heart attacks and strokes. As a result, clinical practice guidelines recommend the initiation of anti-hypertensive medications based on the level of patients' elevated blood pressure and cardiovascular risk levels (D'Agostino et al. 2008).

Despite the widespread use of treatment guidelines, determining the optimal timing and intensity of antihypertensive therapy remains a challenging decision problem (Rabi et al. 2020, Whelton et al. 2018, Williams et al. 2018, Zargoush et al. 2018). Blood pressure evolves dynamically over time and is affected by aging, medication use, and individual physiological factors (AlGhatrif et al. 2013, Law et al. 2009, Najjar et al. 2008, Wilkins et al. 2010). Moreover, treatment decisions involve important trade-offs between reducing future cardiovascular risk and avoiding unnecessary medication exposure and treatment disutility (Lawrence et al. 1996, Tengs and Wallace 2000, Zargoush et al. 2018). These considerations make hypertension management a natural setting for dynamic decision-making under uncertainty (Mason et al. 2014, Zargoush et al. 2018).

There is a growing body of research in the fields of health analytics and operations research that uses a Markov decision process (MDP) approach to evaluate optimal treatment policies for chronic diseases. Zargoush et al. (2018) build a dynamic programming (DP) framework for managing patients with hypertension that will identify optimal medication initiation policies based on patients' blood pressure values as well as the overall cardiovascular risk for these patients. The results of the DP model demonstrate that risk-adjusted treatment policies will outperform the guidline based policies by providing a significantly improved long-term health outcome for patients as measured by quality-adjusted life-years (QALYs).

Existing decision-theoretic models characterize the changing nature of both blood pressure levels and cardiovascular risk over a period of time, yet typically represent vascular health to be a static construct. However, there is clinical research that indicates that one of the best indicators of how much cardiovascular risk an individual has is arterial stiffness as measured by pulse wave velocity (PWV), and that arterial stiffness contributes to the progression of hypertension. Arteries will become increasingly stiff, and hypertension will tend to result in more rapid increases in blood pressure, while antihypertensive pharmaceuticals may become less effective. Incorporating arterial stiffness into the current existing decision models for hypertension treatment in clinical practice has not been done, but it is of clinical significance.

This paper dives into the connection between arterial stiffness and optimal hypertension drug therapy arising from a gap in knowledge. In our dynamic model of arterial stiffness, we can capture how arterial stiffness evolves over time and how it affects both cardiovascular risk and the efficacy of antihypertensive medication. This provides an opportunity to incorporate arterial stiffness into our model and observe some of the documented relationships between vascular aging and the efficacy of antihypertensive medications.

In our study, we integrate 3 components into our model for the assessment of hypertension: (1) Kalman filtering applied to a clinical cohort of patients with hypertension will provide blood pressure dynamics at the level of an individual patient with variability in blood pressure progressions and measurement variability. (2) The progression of arterial stiffness will follow a stochastic process based on a combination of age, blood pressure, and medications. (3) The effectiveness of antihypertensive medications will be attenuated as pulse wave velocity increases in the body, reflecting that stiff arteries respond very little, if at all, to antihypertensive drugs.

We intend to answer the following research questions using our new model:

1. How does incorporating arterial stiffness into the state space impact the optimal antihypertensive treatment policies?

2. What is the relation between vascular aging and the efficacy of a medication in determining treatment thresholds?

3. Relative to the available guideline-based treatment, what are the long-term health benefits associated with stiffness aware treatment guidelines?

To answer the above questions, we employ backward induction to solve the dynamic treatment problem, with the goal of developing optimal treatment policies. The optimal policies are then compared to the guideline-based treatment policies recommended by the British Hypertension Society (BHS). The performance of each treatment policy is evaluated through cohort simulation, that generates patients' health trajectories over time and calculates expected Quality Adjusted Life Years (QALYs) under each policy.

Our analysis presented several major findings. At first, arterial stiffness should be a variable included in the treatment decision-making process, which will provide for a different optimal treatment policy structure. At the moment, the current SBP-based guidelines lead to an earlier initiation of treatment for more patients with elevated PWV than is currently recommended. The inclusion of both the PWV and treatment efficacy will increase long-term outcomes for patients. The stiffness aware policy has a greater expected number of QALYs through all simulated cohorts than would be achieved through the BHS guideline. Finally, the robustness of the findings were demonstrated through the various sensitivity analyses, which show that the conclusions maintain validity even with potential uncertainties related to blood pressure (variance) or with curvatures related to the stiffness attenuation function.

This research adds to both operations management and health services analytics. First, we expand upon existing management decision problems regarding the treatment of hypertension by considering arterial stiffness to be a time-varying physiological state that influences the effectiveness of treatment and risk of cardiovascular disease (CVD). Second, we incorporate dynamic optimization and clinical data to develop estimates of how each patient will progress over time with regard to both his or her disease and response to treatment. Our findings demonstrate that vascular aging biomarkers could potentially be useful for clinical decision support systems as part of their management of hypertension.

2. Literature Review

Our research draws on several streams of literature: clinical risk assessment for cardiovascular disease, dynamic treatment optimisation based on Markov models, and the growing clinical literature concerning arterial stiffness and vascular age.

Firstly, the literature on clinical cardiovascular risk assessment describes the means for developing and employing clinical risk scores within treatment guidelines for patients at risk of developing cardiovascular disease. In both hypertension and cardiovascular disease, a long series of epidemiological studies and clinical trials have established blood pressure as one of the most robust independent predictors of cardiovascular events (D'Agostino et al. 2008). Much of that research has produced risk prediction tools; one of the best known is the Framingham risk score which estimates an individual patient's risk of developing coronary heart disease and stroke. The Framingham risk score is determined using a variety of standard risk factors including blood pressure, total cholesterol, smoking status, and diabetes. The clinical risk models used for these guidelines provide a foundation for care in managing patients with hypertension, while at the same time, these models are referred to in making decisions about the care of the patients being evaluated. The majority of risk assessment models used for making decisions do not take into account that physiologic states are dynamic over time, and most risk assessments provide solely a set of static figures based on present-day patient-related variables.

Dynamic decision models that apply to medical treatment policies represent a second body of literature that focuses on finding optimal strategies for delivering care to patients. In particular, Markov Decision Processes (MDPs) have been successfully utilized to analyze chronic disease management problems which involve making sequential and uncertain treatment choices for patients over time. Examples include treatment policies for HIV therapy Hauskrecht (2000), diabetes management Puterman (1994), and cardiovascular disease prevention. Most closely related to our work, Zargoush et al. (2018) develop a dynamic programming framework for hypertension treatment that determines optimal medication initiation thresholds based on patients' blood pressure levels and cardiovascular risk. Their study demonstrates how treatment policy based on assessment of risk-adjusted treatment policies will bring greater improvements in the patient's outcome than Traditional Physician Guidelines like that provided by the British Hypertension Society (BHS). This framework is important to inform the sequential decision-making process in managing patients with high blood pressure, but it does not include any metrics for vascular aging which is closely associated with CVD risk and treatment response.

The third body of literature surrounds characteristics of arterial stiffness and how they influence health outcomes. Arterial stiffness, commonly assessed through the utilization of Pulse Wave Velocity (PWV), has been shown to be one of the strongest indicators of cardiovascular risk and mortality. (Ben-Shlomo et al. 2014). Numerous clinical studies show that elevated PWV is associated with increased incidence of coronary heart disease, stroke, and other cardiovascular events, even after controlling for traditional risk factors such as blood pressure and cholesterol. In addition, arterial stiffness is closely linked to vascular aging and may influence the progression of hypertension itself (Boutouyrie et al. 2010). Recent studies also suggest that the effectiveness of antihypertensive medications may decline in patients with advanced arterial stiffness, reflecting reduced vascular elasticity.

The connection between three areas of study is our use of the arterial stiffness in a dynamic model of treatment for hypertension. The focus of previous models based on multiple diagnostic parameters (MDP) has been primarily on blood pressure and thus on the resolved cardiovascular risk scores. Our model introduces the use of Pulse Wave Velocity (PWV) as an evolving, dynamic physiological status that can evolve over time and have an impact on both the risks associated with heart disease and the treatment of heart disease. By including vascular aging into the model, it provides a more comprehensive representation of the physiological mechanisms involved in the progression of hypertension and the response to treatment.

3. Main Decision-Theoretic Model

3.1. Decision Epochs

The decision epochs in our problem are denoted by $t = 0, 1, 2, \ldots, T$, where t represents the number of quarter years above the age of 40 (Lovibond et al., 2011). Therefore, t is linked to the patient's age, and $t = 0$ and $t = T$ represent the minimum and maximum ages in our study, respectively.

3.2. State Space

The state space at period t, denoted by x_t, contains all patient's necessary information for decision-making at the visit. The state space consists of three elements: patient's health state, arterial stiffness state, and medication status, denoted by b_t, w_t, and m_t, respectively. Therefore:

$$x_t = (b_t, m_t, w_t)$$

where

$$b_t \in B, \qquad m_t = (m_{1t}, m_{2t}, \ldots, m_{Mt}), \qquad w_t \in W$$

Here, B represents the set of blood pressure health states and W represents the set of arterial stiffness levels. The pair (b_t, w_t) jointly characterizes the patient's physiological condition, while m_t captures the prescribed treatment status.

In what follows, we explain the construction of the feasible sets B, W, and the vector m_t.

3.2.1. Patient's Health State. In our model, we distinguish between the period before and after the first cardiovascular event. Accordingly, we divide the health-state space into two subsets: H and A, where H corresponds to living (event–free) health states and A corresponds to post–event and terminal states.

The patient's health state at each period is represented by $b_t \in B$, where

$$B = H \cup A.$$

The subset H contains discrete blood pressure levels between the minimum $\underline{B}$ and maximum $\overline{B}$ values and represents the recurring non-terminal states (i.e., the patient is alive and has not experienced a major cardiovascular event).

The subset A contains three special states:

$$\{\overline{B}+1\}, \quad \{\overline{B}+2\}, \quad \{\overline{B}+3\},$$

representing coronary heart disease (CHD), stroke (ST), and non–CVD mortality, respectively.

The CHD and ST states represent post–event health states. Once a patient transitions to CHD or ST, the process enters the post–event phase and subsequently transitions to the death state. The non–CVD mortality state $\{\overline{B}+3\}$ is absorbing.

From a modeling perspective, states in H are recurring, CHD and ST are non-recurring post–event states, and $\{\overline{B}+3\}$ is the only absorbing state.

3.2.2. Patient's Medication Status. The vector m_t contains the list of antihypertensive medications prescribed by period t. We represent the patient's medication status by

$$m_t = (m_{1t}, m_{2t}, \ldots, m_{Mt}),$$

where $m_{it} \in \{0, 1\}$ indicates whether the patient is taking medication i in period t, and M is the total number of medications in the problem.

3.2.3. Patient's Arterial Stiffness State. In addition to blood pressure and medication status, we explicitly model arterial stiffness as a physiological state variable. Arterial stiffness is measured by pulse wave velocity (PWV).

We represent the patient's stiffness state at each period by $w_t \in W$, where W is a finite set of discretized PWV levels:

$$W = \{\underline{W}, \underline{W} + \Delta_w, \dots, \overline{W}\}.$$

Unlike the absorbing health states, arterial stiffness is a recurring condition. Therefore, all elements of W correspond to non-absorbing states.

3.3. Action Space

At each decision epoch t, the decision-maker selects an action

$$a_t \in A_t(x_t),$$

where the feasible action set is defined as

$$A_t(x_t) = \{0\} \cup I_t.$$

Here, 0 denotes maintaining the current regimen (i.e., no new medication is prescribed), and $i \in I_t$ denotes initiating antihypertensive drug class i.

The set I_t represents the eligible medications that have not been prescribed before period t, namely

$$I_t = \{\, i : m_{it} = 0 \,\}.$$

Thus, at each decision epoch, the decision-maker either prescribes one of the eligible medications or does nothing. For notational convenience, we suppress the dependence of the action on the state and write a_t instead of $a_t(x_t)$ in the remainder of the paper.

The feasible action set depends on the patient state (b_t, w_t, m_t) through the eligibility set I_t.

3.4. State Transitions

3.4.1. Blood Pressure Transitions. In this model, systolic blood pressure (SBP) evolves as a latent continuous physiological state. Let $\tilde{b}_t$ denote the latent true SBP at time t. The SBP transition used in the decision model is

$$\tilde{b}_{t+1} = \tilde{b}_t - \sum_i \varepsilon_i(\tilde{b}_t)\, \beta(w_t)\, \mathbf{1}_{a_t=i} + \gamma\Delta t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2\Delta t). \tag{1}$$

The components of the transition are defined as follows:

- $\varepsilon_i(\tilde{b}_t)$ is the class-specific SBP reduction associated with initiating medication class i at baseline SBP level $\tilde{b}_t$.
- $\mathbf{1}_{a_t=i}$ is an indicator function equal to 1 if medication class i is initiated at time t, and 0 otherwise.
- γ represents the long-run deterministic drift in SBP due to aging.
- η_t is a stochastic innovation term capturing short-term physiological fluctuations in latent SBP.
- σ^2 is the intrinsic process variance per unit time.

The variance scales linearly with Δt, consistent with a continuous-time diffusion approximation of blood pressure dynamics.

Arterial stiffness affects SBP only through treatment responsiveness. It does not reverse the direction of treatment effect; antihypertensive therapy still lowers SBP, but the magnitude of reduction may be smaller for patients with elevated pulse wave velocity (PWV) (Fan et al. 2020, Protogerou et al. 2009).

Medication Effectiveness (ε_i). Following the meta-analysis of Law et al. (2009), we adopt the standard-dose class-specific effectiveness functions used in hypertension policy modeling. For baseline SBP level $\tilde{b}_t$, the treatment effects are

$$\varepsilon_{\text{ACEI}}(\tilde{b}_t) = 0.0347\,\tilde{b}_t, \tag{2}$$

$$\varepsilon_{\text{DU}}(\tilde{b}_t) = 0.0360\,\tilde{b}_t, \tag{3}$$

$$\varepsilon_{\text{CCB}}(\tilde{b}_t) = 0.0364\,\tilde{b}_t, \tag{4}$$

$$\varepsilon_{\text{BB}}(\tilde{b}_t) = 0.0373\,\tilde{b}_t, \tag{5}$$

$$\varepsilon_{\text{ARB}}(\tilde{b}_t) = 0.0424\,\tilde{b}_t. \tag{6}$$

These represent baseline clinical treatment effects and are kept unchanged to preserve comparability with established hypertension decision-analytic models.

Accordingly, the ordering of medication effectiveness is

$$\text{ACEI} < \text{DU} < \text{CCB} < \text{BB} < \text{ARB}.$$

Gender-Specific Drift (γ). The long-run evolution of SBP includes a deterministic aging component. Following prior hypertension decision models, the baseline specification uses gender-specific population estimates derived from Canadian longitudinal epidemiological data (Wilkins et al. 2010):

$$\gamma_{\text{male}} = 0.30 \text{ mmHg/year}, \qquad \gamma_{\text{female}} = 0.37 \text{ mmHg/year}.$$

These parameters represent secular physiological progression independent of treatment decisions and serve as the primary disease-progression assumption in the baseline model.

External Plausibility Check. Using the same state-space framework described below, we also estimated drift from treatment-stable visit intervals in the MGH cohort. The resulting estimates were

$$\hat{\gamma}_{\text{female}} = 0.939 \text{ mmHg/year}, \qquad \hat{\gamma}_{\text{male}} = 0.723 \text{ mmHg/year}.$$

Because the cohort consists of treated hypertensive patients under specialist follow-up, these values reflect residual progression under clinical care rather than natural aging alone. Therefore, cohort-derived values are not used in the baseline decision model but are reported to demonstrate external plausibility of the SBP state dynamics.

Robustness: Stiffness-Dependent Progression. Although the baseline model assumes arterial stiffness influences SBP only through treatment responsiveness, longitudinal evidence suggests that vascular stiffness may also accelerate long-term SBP progression. To evaluate this mechanism, we consider an alternative specification

$$\gamma(w) = \gamma_0 + \gamma_1(w - w_{\text{ref}}),$$

and evaluate clinically supported ranges

$$\gamma_1 \in [0, 0.15] \text{ mmHg/year per m/s},$$

as well as an upper-bound scenario $\gamma_1 = 0.22$ mmHg/year per m/s reported in Najjar et al. (2008).

The baseline specification corresponds to $\gamma_1 = 0$, ensuring that arterial stiffness affects treatment response but not the underlying natural history of SBP.

Variance of the SBP Transition Process (σ^2). To quantify stochastic uncertainty in SBP evolution, we estimate the latent transition variance using repeated measurements from the MGH hypertension cohort. Patients with fewer than three visits are excluded, and observations collected during medication changes are removed to retain treatment-stable intervals.

Table 1 summarizes the baseline characteristics of the MGH cohort used for variance estimation.

Table 1 Baseline Characteristics for the MGH Hypertension Cohort

Factor	Value
Total number of patients	84
Male	42 (50%)
Female	42 (50%)
Minimum age	32
Maximum age	87
Minimum number of visits	1
Maximum number of visits	15
Mean number of visits	5.16
Median number of visits	4.0
Total number of visits	387
Earliest visit	Feb. 2001
Latest visit	Feb. 2026
Valid patients ($\geq$ 3 visits)	57 (68%)
Valid male patients	29 (51%)
Valid female patients	28 (49%)
Total number of valid visits	358

Clinic SBP measurements contain substantial measurement error and are observed at irregular intervals. Therefore, raw visit-to-visit differences do not directly represent physiological variability. Instead, we estimate the latent SBP trajectory using a linear Gaussian state-space model (Kalman filter):

$$\tilde{b}_{t+1} = \tilde{b}_t + \gamma \Delta t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2), \tag{7}$$

$$b_t^{\text{obs}} = \tilde{b}_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, r), \tag{8}$$

where $\tilde{b}_t$ denotes true latent SBP, σ^2 is the intrinsic physiological process variance, and r represents measurement error.

Parameters are estimated by maximum likelihood separately by gender. For the 3-month decision epoch used in the MDP, the estimated transition variances are

$$\sigma^2_{\text{female}} = 5.73 \text{ mmHg}^2, \qquad \sigma^2_{\text{male}} = 3.94 \text{ mmHg}^2.$$

These correspond to latent transition standard deviations

$$\sigma_{\text{female}} = 2.39 \text{ mmHg}, \qquad \sigma_{\text{male}} = 1.99 \text{ mmHg}.$$

The process variance σ^2 is assumed independent of arterial stiffness. Stiffness modifies treatment responsiveness but does not alter short-term physiological volatility.

Calibration of the Stiffness Attenuation Function ($\beta(w)$). Arterial stiffness modifies realized treatment impact through a monotone attenuation factor

$$\beta(w) = \frac{w_{\max} - w}{w_{\max} - w_{\text{ref}}(age)}, \tag{9}$$

where w denotes pulse wave velocity (PWV), $w_{\text{ref}}(age)$ is the age-adjusted reference stiffness level, and $w_{\max}$ is the empirical 95th percentile of PWV in the cohort.

This specification satisfies

$$\beta(w_{\text{ref}}) = 1, \qquad \beta(w_{\max}) = 0,$$

so that patients with normal vascular compliance experience the full medication effect, while extremely stiff arteries approach minimal treatment responsiveness.

The effective treatment impact entering the SBP transition is

$$\varepsilon_i^{\text{eff}}(\tilde{b}_t, w_t) = \varepsilon_i(\tilde{b}_t)\,\beta(w_t).$$

Because stiffness scales all medication classes proportionally, the relative ordering of medication effectiveness remains unchanged.

PWV age-stratified reference values (w_{ref}). We calibrate the age-adjusted reference stiffness $w_{\text{ref}}(age)$ using normative carotid–femoral PWV values from Table 5 of Boutouyrie et al. (2010). The upper reference values correspond to age-adjusted thresholds commonly used to characterize increased arterial stiffness.

Table 2 **Age-stratified reference values for carotid–femoral PWV (w_{ref}).**

Age group (years)	Mean PWV (m/s)	Upper reference (mean + 2SD) (m/s)
< 30	6.1	7.5
30–39	6.6	8.9
40–49	7.0	9.6
50–59	7.6	10.5
60–69	9.1	12.9
≥ 70	10.4	15.6

In our model, for a patient whose baseline age falls into age group g, we define $w_{\mathrm{ref}}(g)$ as the mean PWV in Table 2. As a conservative alternative in sensitivity analyses, we may use the upper-normal reference level (mean + 2SD) for the same age group.

PWV distribution and maximum stiffness value ($w_{\max}$). We set the maximum stiffness value $w_{\max}$ to the empirical 95th percentile of observed PWV in the MGH cohort to capture the clinically observed upper tail while limiting sensitivity to extreme values.

Table 3 **PWV distribution of the MGH cohort and choice of $w_{\max}$.**

Statistic	PWV (m/s)
Mean	9.29
Median	9.25
25th percentile	8.13
75th percentile	10.57
95th percentile ($w_{\max}$)	11.889
Maximum	15.1

Finally, stiffness modifies realized SBP reductions through the effective treatment impact

$$\varepsilon_i^{\mathrm{eff}}(b_t, w_t) = \varepsilon_i(b_t)\,\beta(w_t), \qquad \beta(w_t) \in [0, 1], \tag{10}$$

where $\beta(\cdot)$ is nonnegative and nonincreasing in PWV.

3.4.2. Arterial Stiffness Transitions. In addition to blood pressure dynamics, we model arterial stiffness, measured by pulse wave velocity (PWV), as a slowly evolving physiological state. A large body of clinical evidence indicates that arterial stiffness increases with aging and elevated blood pressure (Benetos et al. 2002, Boutouyrie et al. 2010). Furthermore, antihypertensive therapies: particularly those targeting the renin, angiotensin system, and calcium channel blockers have been shown to reduce arterial stiffness beyond their immediate blood pressure lowering effect (London and Pannier 2004, Protogerou et al. 2009). Arterial stiffness is also an independent predictor of future cardiovascular events (Mitchell et al. 2010).

Motivated by this evidence, we model stiffness evolution as a gradual progression process driven by aging, hemodynamic load, and treatment effects. Let w_t denote arterial stiffness at time t. The transition is defined as

$$w_{t+1} = w_t + \alpha\Delta t + \theta_w(b_t - b^{\mathrm{ref}})\Delta t - \sum_i \phi_i \mathbf{1}_{a_t=i} + \nu_t, \tag{11}$$

where Δt denotes the decision interval length, b_t is systolic blood pressure (SBP), and $\nu_t \sim \mathcal{N}(0, \sigma_w^2)$ captures unobserved physiological variability.

The components of (11) have the following interpretations:

- $\alpha\Delta t$ represents the intrinsic aging process of the arterial wall. Even in the absence of hypertension, elastic fibers progressively degenerate and are replaced by collagen, leading to a gradual increase in pulse wave velocity over time (Boutouyrie et al. 2010). We estimated the alpha values from our MGH cohort based on gender and age using the regression reference computed by AlGhatrif et al. (2013). The corresponding values of our cohort are:

Table 4 **Age-bin intrinsic stiffening rate $\alpha(\text{age})$ (m/s/year) for men and women implied by BLSA mixed-effects time-interaction estimates. Age is evaluated at bin midpoints with $A = \text{age}/10$.**

Age group	Midpoint	α_{men}	α_{women}
< 40	35	0.00075	-0.02550
40–50	45	0.00575	0.00350
50–60	55	0.03675	0.03250
60–70	65	0.09375	0.06150
70–80	75	0.17675	0.09050
≥ 80	85	0.28575	0.11950

Based on longitudinal data from young adults with youth-onset type 2 diabetes, carotid–femoral pulse wave velocity increased by approximately 0.15 m/s per year, which we used it to for diabetic patients as the aging parameter α in our stiffness transition model (Shah et al. 2022).

- $\theta_w(b_t - b^{\mathrm{ref}})\Delta t$ represents pressure–induced structural damage. When systolic blood pressure exceeds the normal level b^{ref}, the artery is exposed to excess mechanical load, which accelerates vascular remodeling and increases stiffness. Thus, patients with higher sustained blood pressure experience faster stiffening than normal individuals (Benetos et al. 2002, Boutouyrie et al. 2010). We set $b^{\text{ref}} = 120$ mmHg, corresponding to normotensive blood pressure levels defined in major hypertension guidelines across the globe (Rabi et al. 2020, Whelton et al. 2018, Williams et al. 2018).

• ϕ_i represents treatment–induced vascular improvement associated with initiating drug class i. Certain antihypertensive therapies not only lower blood pressure but also improve arterial compliance through biological mechanisms such as reduced collagen deposition and improved endothelial function (London and Pannier 2004, Protogerou et al. 2009).

The percentage affect of medications on PVW has been found from numerious studies, sumamrized below as ϕ_i values for our work:

Table 5 Effect of Antihypertensive Drug Classes on Pulse Wave Velocity (PWV)

Medication Class	Effect on PWV	Treatment Duration	Reference
Beta-blockers (BB)	↓ ~1.12 m/s (~10–12%)	2–6 months (pooled trials)	(Niu and Qi 2016)
Calcium Channel Blockers (CCB)	↓ ~1.7 m/s (~15–17%)	38 weeks (9 months)	(Hayoz et al. 2012)
Angiotensin Receptor Blockers (ARB)	↓ 0.43 m/s	2–12 months (pooled trials)	(Peng et al. 2015)
ACE Inhibitors (ACEI)	↓ ~1.15 m/s (9–10%)	3–12 months (pooled trials)	(Mallareddy et al. 2006)
Thiazide(-like) Diuretics (DU)	↓ ~1.18 m/s (chlorthalidone)	8 weeks (2 months)	(Kwon et al. 2013)

• ν_t captures residual physiological variability in arterial stiffness progression not explained by age, blood pressure, or treatment.

Because our cohort contains only a single cfPWV measurement per individual, we calibrate the annual stiffness innovation variance using longitudinal cfPWV dispersion reported in the ELSA–Brasil cohort (Baldo et al. 2018). In a healthy reference subset, the annual cfPWV progression was reported as:

$$0.0504 \pm 0.153 \text{ m/s/year (men)}, \qquad 0.0606 \pm 0.139 \text{ m/s/year (women)}.$$

We therefore model the yearly innovation term as:

$$\nu_t \sim \mathcal{N}(0, \sigma^2_{w,\text{yr}} \Delta t),$$

with sex-specific annual variances

$$\sigma^2_{w,\text{yr,men}} = (0.153)^2 = 0.0234, \qquad \sigma^2_{w,\text{yr,women}} = (0.139)^2 = 0.0193,$$

where Δt is measured in years.

3.4.3. Medication treatment transitions. The transition between medication states is deterministic and simply tracks whether a medication has been initiated. Once a medication is prescribed it cannot be withdrawn, and at most one new medication can be added at each decision epoch. The medication state therefore evolves as

$$m_{t+1,i} = \begin{cases} 1, & \text{if } a_t = i, \\ m_{t,i}, & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, M. \tag{12}$$

with $a_t = 0$ corresponding to no action taken, in which case $m_{t+1} = m_t$.

3.4.4. Cardiovascular Event Risk and Health-State Transitions. $B = H \cup A$ is the health-state space consisting of transient living states H and absorbing states $A = \{\overline{B}+1, \overline{B}+2, \overline{B}+3\}$ corresponding to coronary heart disease (CHD), stroke (ST), and non–CVD mortality, respectively. The arterial stiffness level $w_t \in W$ is an additional physiological state, while the medication vector m_t evolves according to (12).

Following Zargoush et al. (2018), at each decision epoch the patient either remains alive and the physiological state evolves, or transitions to one of the absorbing health states. Cardiovascular risk is evaluated using the Framingham risk equations (D'Agostino et al. 2008). We retain this structure and extend the risk index to depend on arterial stiffness in addition to systolic blood pressure, consistent with evidence that pulse wave velocity (PWV) provides prognostic information beyond blood pressure levels (Mitchell et al. 2010, Vlachopoulos et al. 2010).

3.5. Probability of Nonfatal Cardiovascular Events

3.5.1. Probability of Nonfatal Events We follow the formulation of Zargoush et al. (2018) and estimate nonfatal cardiovascular event risks using the Framingham risk equations reported in D'Agostino et al. (2008). The Framingham model provides gender-specific risk indices in which cardiovascular risk is a function of age t, total cholesterol (CLT), high-density lipoprotein cholesterol (HDL), systolic blood pressure (SBP), smoking status (SMK), and diabetes status (DIAB). However, epidemiological evidence suggests that carotid–femoral pulse wave velocity (PWV) independently predicts cardiovascular events even after adjusting for SBP (Ben-Shlomo et al. 2014). To incorporate arterial stiffness while avoiding implicit double-counting of age effects already embedded in the Framingham model, we introduce PWV through its proportional deviation from an age-adjusted reference level.

For gender $h \in 1, 2$ (male, female), we define the Framingham risk index as:

$$\theta_t^h(x_t) = \exp\left(\beta_1^h \ln(t) + \beta_2^h \ln(\mathrm{CLT}) - \beta_3^h \ln(\mathrm{HDL}) + \beta_4^h \ln(b_t) + \beta_5^h \mathrm{SMK} + \beta_6^h \mathrm{DIAB} + \beta_7 \ln\left(\frac{w_t}{w_{\mathrm{ref}}(t)}\right) + \beta_0^h\right) \tag{13}$$

where b_t denotes systolic blood pressure, w_t denotes carotid–femoral pulse wave velocity (PWV), and $w_{\mathrm{ref}}(t)$ represents the age-specific reference PWV derived from population norms. Under this formulation, stiffness increases cardiovascular risk only when it exceeds the level expected for a given age.

The coefficient on the PWV term is motivated by meta-analytic evidence showing that a 1 m/s increase in PWV is associated with approximately a 14–15 percent increase in cardiovascular event risk (Vlachopoulos et al. 2010). Under the logarithmic specification, an increase in PWV from w to $w+1$ multiplies the risk index by $\left(\frac{w+1}{w}\right)^{\beta_7}$, which corresponds approximately to a 14–15 percent increase over clinically relevant PWV ranges. Accordingly, we set

$$\beta_7 = \ln(1.15) \approx 0.1398.$$

Gender-specific coefficients are:

$$\begin{aligned}
\textbf{Male } (h=1)\textbf{:} \quad & \beta_1^1 = 3.06117, \quad \beta_2^1 = 1.12370, \quad \beta_3^1 = 0.93263, \\
& \beta_4^1 = 1.93303, \quad \beta_5^1 = 0.65451, \quad \beta_6^1 = 0.57367, \quad \beta_0^1 = -23.9802; \\
\textbf{Female } (h=2)\textbf{:} \quad & \beta_1^2 = 2.3288, \quad \beta_2^2 = 1.20904, \quad \beta_3^2 = 0.70833, \\
& \beta_4^2 = 2.67157, \quad \beta_5^2 = 0.52873, \quad \beta_6^2 = 0.69154, \quad \beta_0^2 = -26.1931.
\end{aligned}$$

The probability of nonfatal cardiovascular event $j \in \{1, 2\}$ (CHD, ST) for gender h is:

$$\alpha_{j,t}^h(x_t) = k_j^h \left(1 - c_h^{\theta_t^h(x_t)}\right), \qquad j \in \{1, 2\}, \tag{14}$$

where calibration constants k_j^h and c_h are given by:

	$j = 1$ (CHD)	$j = 2$ (ST)
$h = 1$	0.7174	0.1590
$h = 2$	0.6086	0.2385

$$c_1 = 0.88936, \qquad c_2 = 0.95012.$$

3.5.2. Non–CVD mortality: Following Lewington et al. (2002), we model the probability of non–CVD mortality over decision interval t as an age-dependent exponential function:

$$\pi_{3,t}(age_t) = 5 \times 10^{-5} \exp(0.0824\, age_t), \tag{15}$$

which provides an excellent fit to the reported curve ($R^2 = 0.9988$).

To incorporate evidence that arterial stiffness predicts non–CVD mortality, we perform a secondary specification using results from a post hoc secondary analysis of SPRINT, which reported that each 1 SD increase in estimated PWV (SD = 1.7 m/s) was associated with a hazard ratio of 1.76 for death from non–cardiovascular causes (Vlachopoulos et al. 2019). We translate this association to a log-linear multiplicative adjustment:

$$\pi_{3,t}(x_t) = 5 \times 10^{-5} \exp(0.0824\, age_t) \exp\big(\theta_{\text{PWV}} \ln(w_t)\big), \qquad \theta_{\text{PWV}} = \frac{\ln(1.76)}{1.7}. \tag{16}$$

3.5.3. Joint transition kernel. If a terminal event $j \in \{1, 2, 3\}$ occurs during interval t, the post–event health state becomes $b_{t+1} = \overline{B} + j$; since the process terminates, we keep $(m_{t+1}, w_{t+1}) = (m_t, w_t)$ for notational completeness.

Correlated physiological evolution. For $b_t \in H$, the joint evolution of latent systolic blood pressure $\tilde{b}_{t+1}$ and arterial stiffness w_{t+1} is modeled as a bivariate Gaussian transition:

$$\begin{pmatrix} \tilde{b}_{t+1} \\ w_{t+1} \end{pmatrix} \Bigg| (x_t, a_t, h) \sim \mathcal{N}\big(\mu_t(x_t, a_t),\ \Sigma_h(\Delta t)\big),$$

where the conditional mean vector is

$$\mu_t(x_t, a_t) = \begin{pmatrix} \tilde{b}_t - \sum_i \varepsilon_i(\tilde{b}_t)\beta(w_t)\mathbf{1}_{a_t=i} + \gamma\Delta t \\ w_t + \alpha\Delta t + \theta_w(b_t - b^{\text{ref}})\Delta t - \sum_i \phi_i \mathbf{1}_{a_t=i} \end{pmatrix}.$$

We allow correlated innovations between SBP and PWV to capture shared unobserved physiological drivers. The covariance matrix is

$$\Sigma_h(\Delta t) = \Delta t \begin{pmatrix} \sigma_h^2 & \rho\, \sigma_h \sigma_{w,h,\text{yr}} \\ \rho\, \sigma_h \sigma_{w,h,\text{yr}} & \sigma^2_{w,h,\text{yr}} \end{pmatrix}, \qquad -1 < \rho < 1,$$

where σ_h^2 is the latent SBP process variance, $\sigma^2_{w,h,\text{yr}}$ is the annual PWV innovation variance, and Δt is measured in years. Here, $h \in \{1, 2\}$ denote gender (male, female), which determines gender-specific SBP and PWV innovation variances.

In the baseline specification we set

$$\rho = 0.22,$$

based on the empirical correlation between observed SBP and PWV in the MGH cohort. Sensitivity analysis evaluates $\rho \in [0, 0.4]$.

Quarterly calibrated covariance. We use a quarterly decision interval $\Delta t = 0.25$ years. SBP innovation variance is estimated directly at the 3-month interval. PWV innovation variance is annual and scaled by Δt. The SBP–PWV innovation correlation is set to $\rho = 0.2203$.

Accordingly, the joint quarterly covariance matrices are:

Women:

$$\Sigma_{\text{female}} = \begin{pmatrix} 5.73 & 0.0366 \\ 0.0366 & 0.004825 \end{pmatrix}.$$

Men:

$$\Sigma_{\text{male}} = \begin{pmatrix} 3.94 & 0.0334 \\ 0.0334 & 0.00585 \end{pmatrix}.$$

$$(\tilde{b}_{t+1}, w_{t+1}) \mid (x_t, a_t, h) \sim \mathcal{N}(\mu_t(x_t, a_t), \Sigma_h).$$

The continuous Gaussian transition is discretized over the finite state grid $H \times W$ to obtain the probability mass function $\tilde{p}_t(b_{t+1}, w_{t+1} \mid x_t, a_t, h)$ used in the Markov kernel.

Given the probability of death $\alpha_{3t}(x_t, a_t)$, the probabilities of facing $\{\overline{B} + j\}$ (with $j = 1$ corresponding to CHD, and $j = 2$ corresponding to ST) can be formulated as

$$\alpha'_{jt}(x_t, a_t) = \big(1 - \alpha_{3t}(x_t, a_t)\big)\alpha_{jt}(x_t, a_t), \qquad j = \{1, 2\}.$$

We can now fully characterize the transition probability function as follows:

$$P_t(x_{t+1} \mid x_t, a_t) = P_t(b_{t+1}, m_{t+1}, w_{t+1} \mid b_t, m_t, w_t, a_t).$$

$$= \begin{cases} \left(1 - \sum_{j=1}^{3} \alpha'_{jt}(x_t, a_t)\right) \tilde{p}_t(b_{t+1}, w_{t+1} \mid x_t, a_t, h), & b_t \in H,\ b_{t+1} \in H, \\ \alpha'_{jt}(x_t, a_t), & b_t \in H,\ b_{t+1} = \{\overline{B} + j\},\ j = \{1, 2, 3\}, \\ 1, & b_t \in A,\ b_{t+1} = \{\overline{B} + 3\}, \\ 0, & \text{otherwise}. \end{cases}$$

with $\alpha'_{3t}(x_t, a_t) = \alpha_{3t}(x_t, a_t)$.

3.6. Reward Function

Our model includes two reward functions for the resulting MDP: one for the immediate rewards of the non-terminal states, and one for the terminal rewards. In the following paragraphs, we describe the construction of these functions.

3.6.1. Immediate Reward Function: Terminal events. Let $j \in \{1, 2, 3\}$ denote CHD, stroke, and non–CVD death, respectively. Given state $x_t = (b_t, m_t, w_t)$ and action a_t, let $\alpha'_{jt}(x_t, a_t)$ denote the probability that terminal event j occurs during period $(t, t+1]$. These probabilities are defined in the transition model and play the same role as in Zargoush et al. (2018). The presence of arterial stiffness affects rewards only through these event probabilities.

Event-free reward. While alive, the patient accrues quality-adjusted life years (QALYs). Let Δt denote the period length (years), and let $d_i \geq 0$ represent the disutility associated with drug class i. With post-decision medication vector $m_{t+1} = f(m_t, a_t)$, define the one-period reward as

$$\hat{r}_t(x_t, a_t) = \Delta t \left(1 - \sum_i d_i \, m_{t+1,i}\right), \qquad m_{t+1} = f(m_t, a_t). \tag{17}$$

Within-period events. We need, however, to account for the possibility of encountering terminal events by the next period. To do so, we prorate the above QALYs according to when the patient might face a terminal event. We assume the events will occur in the middle of the period (Sonnenberg and Beck 1993). In other words, if the event occurs, the patient lives without the event for half interval length, and with the event for the remaining half. Otherwise, the patient lives event-free for the entire period.

Let ω_j be the QALY of a patient who experiences event $\{\overline{B}+j\}$ during period t, where $j \in \{1, 2, 3\}$ and $\omega_3 = 0$, as $j = 3$ corresponds to death. Then, the expected QALY for the entire period can be computed as follows:

$$Q(\overline{B} + j) = \frac{1}{2}\left(\hat{r}_t(x_t, a_t) + \omega_j\right). \tag{18}$$

We can now present the expected immediate reward function as follows:

$$r_t(x_t, a_t) = \left(1 - \sum_{j=1}^{3} \alpha'_{jt}(x_t, a_t)\right) \hat{r}_t(x_t, a_t) + \sum_{j=1}^{3} \alpha'_{jt}(x_t, a_t) Q(\overline{B} + j). \tag{19}$$

The above expression implies that

$$r_t(x_t, a_t) = \hat{r}_t(x_t, a_t)$$

in the absence of any terminal event (i.e., when $\alpha'_{1t} = \alpha'_{2t} = \alpha'_{3t} = 0$).

3.6.2. Lump-Sum Terminal Reward Function: We denote by R_{jt}, the total expected terminal reward for a patient who is in the terminal state $\{\bar{B} + j\}$ in period t. In other words:

$$r_t(\{\bar{B} + j\}, w_t, m_t, a_t) = R_{jt}, \tag{8}$$

where R_{jt} is the quality-adjusted life expectancy (QALE) of a patient who experiences his first non-fatal event or death in period t. In other words, R_{jt} captures both the quality and quantity of the expected life after facing event j, hence $R_{3t} = 0$.

Post-event life expectancy. For nonfatal events $j \in \{1, 2\}$ (CHD and stroke), we define the terminal reward as the expected remaining QALE after the event. Let $h \in \{1, 2\}$ denote gender (male, female). Following Hannerz and Nielsen (2001), gender-specific life expectancy after a cardiovascular event is estimated as

$$LE_t^h = \begin{cases} 0.0073t^2 - 1.3673t + 68.683, & h = 1, \\ 0.0073t^2 - 1.4696t + 76.425, & h = 2, \end{cases} \tag{20}$$

where t denotes age at the time of the event. The fitted curves provide excellent approximation accuracy ($R^2 = 0.9994$ for males; $R^2 = 0.9999$ for females). These equations imply that females experience longer life expectancy following cardiovascular events.

Quality adjustment. Let ω_j denote the QALY weight associated with post-event health state j. Then the baseline terminal reward for gender h is

$$R_{jt}^h = \omega_j \, LE_t^h, \qquad j \in \{1, 2\}. \tag{21}$$

For non–CVD death ($j = 3$), we set

$$R_{3t} = 0.$$

PWV-adjusted life expectancy. Evidence suggests that higher arterial stiffness predicts increased cardiovascular and all-cause mortality even after adjustment for traditional risk factors. In particular, Shokawa et al. (2005) report that individuals with PWV > 9.9 m/s exhibit significantly higher cardiovascular mortality (RR ≈ 4.24) and increased all-cause mortality (RR ≈ 1.42).

To reflect this elevated mortality risk, we adjust post-event life expectancy multiplicatively. Let

$$RR(w_t) = \begin{cases} 1.42, & w_t > 9.9 \text{ m/s}, \\ 1, & \text{otherwise}. \end{cases}$$

The PWV-adjusted life expectancy becomes

$$\widetilde{LE}_t^h = \frac{LE_t^h}{RR(w_t)}. \tag{22}$$

The resulting stiffness-adjusted terminal reward is

$$R_{jt}^h = \omega_j \, \widetilde{LE}_t^h, \qquad j \in \{1, 2\}. \tag{23}$$

This proportional scaling provides a tractable approximation of increased post-event mortality under elevated arterial stiffness

3.7. The MDP Formulation

Putting all the above pieces together, we can now formulate the value function for HTNMP with arterial stiffness. Let

$$v_t(\hat{b}_t, m_t, w_t)$$

be the maximum total discounted expected QALYs earned by a patient given health state b_t, medication state m_t, and arterial stiffness state w_t in period t. The Bellman optimality equations for the non-terminal health states can be written as:

$$\begin{aligned} v_t(\hat{b}_t, m_t, w_t) = \max_{a_t \in A_t(\hat{b}_t, m_t, w_t)} \Bigg\{ & r_t(\hat{b}_t, m_t, w_t, a_t) + \lambda^{\Delta t} \sum_{\hat{b}_{t+1} \in B} \sum_{w_{t+1} \in W} p_t(\hat{b}_{t+1}, w_{t+1} \mid \hat{b}_t, w_t, a_t) \\ & \times v_{t+1}(\hat{b}_{t+1}, m_{t+1}, w_{t+1}) \Bigg\}, \quad \forall t = 0, 1, \ldots, T-1, \end{aligned} \tag{9}$$

where $\lambda \in [0, 1]$ denotes the annual discount factor and $m_{t+1} = f(m_t, a_t)$.

For terminal states $\tilde{b}_t \in \{\bar{B} + j\}$ and terminal decision period T, we define the terminal value functions as follows:

$$v_t(\{\bar{B} + j\}, m_t, w_t) = R_{jt}, \qquad j \in \{1, 2, 3\},\ \forall t = 0, 1, \ldots, T-1, \tag{24}$$

and

$$v_T(b_T, m_T, w_T) = R_T. \tag{10}$$

where R_T is the remaining quality-adjusted life expectancy of a patient at the end of the decision horizon T. We assume $R_T = 0$ when a patient passes the age of 100 in our decision process, as hypertensive patients rarely reach this age (Franco et al. 2005).

4. Structure of Optimal Policies

4.1. Definition of Optimal Policies

The finite-horizon Markov decision process was numerically solved, using backward induction across ages 40 to 100 with treatment and business decisions made every quarter. The goal of this section is to characterize optimal treatment policies in terms of thresholds of systolic blood pressure (SBP) that will trigger or escalate treatment.

To evaluate the value of arterial stiffness as an information source, we develop three models that vary from each other only with respect to the way in which arterial stiffness (approximate size of the arterial wall's resistance and/or PWV) is used to influence the treatment decision process. The three models have been created to isolate the contribution of vascular based information, whilst keeping the decision structure for medications the same.

To achieve this we are comparing three models which are identical in every aspect, but differ only in the manner in which the patient is assessed using the underlying arterial stiffness value (PWV): a latent PWV benchmark model in which there is an assumed progressive change of PWV to provide measurement of stiffness - but the change in PWV is not assessed in making treatment decisions; reference-PWV model that uses a proxy value based upon the patient's chronological age to assess the level of stiffness; and, a fully observed PWV model that uses the patient's actual size of PWV in making treatment decisions.

Table 6 provides a summary of the various PWV information structures employed by the three models used in the decision making process.

Table 6 **PWV information structures used to evaluate the value of observing arterial stiffness.**

Model	SBP observed	PWV observed
Latent-PWV benchmark (NI)	✓	⊠
Reference-PWV proxy model (PI)	✓	⊠
Fully observed PWV model (FI)	✓	✓

In all three models the SBP and PWV of the patient are subject to stochastic (random) variation within the population under study; the latent PWV benchmark model ignores the impact of PWV on the decision making rules; the reference-PWV model uses a proxy ($w_{\text{ref}}(\text{age})$) stiffness value as an alternative to determining the PWV, while the fully observed PWV model includes the true PWV value in the treatment decision process.

Latent-PWV Benchmark (NI) Model The Latent-PWV Benchmark (NI) Model uses an individual's arterial stiffness based on individual patient physiology, but it cannot be determined by the decision maker. The decision rule uses only observable clinical conditions (such as systolic blood pressure and prescribed medication). Thus, a decision rule takes the form of $\pi(b, m, t)$, in that while PWV will have an effect on future cardiovascular risk because of transitions, it is not directly used to determine the appropriate treatment. This consideration reflects the common type of information structure used to guide and manage hypertension according to standard clinical guidelines, since vascular stiffness is not typically the main determinant of appropriate treatment.

Reference-PWV Proxy (PI) Model Partial information (PI) models utilize an estimate of stiffness $w_{\text{ref}}(\text{age})$ based on a patient's age as a surrogate for the unobservable value of PWV. This approach describes the clinical practice of approximating vascular aging by using age. A decision rule with this model uses the Stiffness Associated with Age, w_{ref}; thus, Decision Policy will take the form $\pi(b, m, w_{\text{ref}}(\text{age}))$. This model provides some agreement with aging of vascular systems; however, there is no consideration of how the patient is different from the expected or average rate of stiffening.

Fully observed PWV (FI) Model This FI model captures the reality of clinical decision making in a fully informed patient and provider context where the true PWV state is known and included directly in the decision making of treatment. The decision rule is determined based on the complete physiological state of the patient ($\pi(b, m, w)$) and can change the threshold for initiating treatment according to the patient's current level of arterial stiffness. The FI model demonstrates the clinical benefit of using direct measurement of vascular stiffness for decision making in the management of hypertension.

In order to visualize the policies developed for policy C, which depend on three state variables (b, m, w), we need to take a two-dimensional slice through the state space. Therefore, to evaluate the policies visually, we average them along the reference vascular aging path ($w = w_{\text{ref}}(\text{age})$), which represents how arterial stiffness develops over the course of a patient's life, and display the optimal treatment recommendations based on SBP at various clinically relevant levels of blood pressure while keeping the PWV constant at the age-related reference value at the time of making that recommendation.

4.2. Policy Comparison Across PWV Information Models

We now compare the optimal treshold policies generated under the three PWV information structures for a risk-free male, for short for a man who does not smoke and does not have diabetes.

For each profile we display the optimal medication prescription models obtained under NI (latent PWV), PI (reference-PWV proxy), and FI (fully observed PWV).

In each heatmap, the horizontal axis represents age and the vertical axis represents systolic blood pressure (SBP). White regions denote *Do Nothing*, while coloured regions correspond to initiating a specific medication class. For a fixed age, treatment intensity increases with SBP, producing a threshold structure where treatment is initiated once SBP exceeds a critical level.

Risk-free male. Figure 1 shows the optimal medication prescription when PWV is ignored in the decision-making where at 2 it is for prescribing medication by current regimen state of the patient.

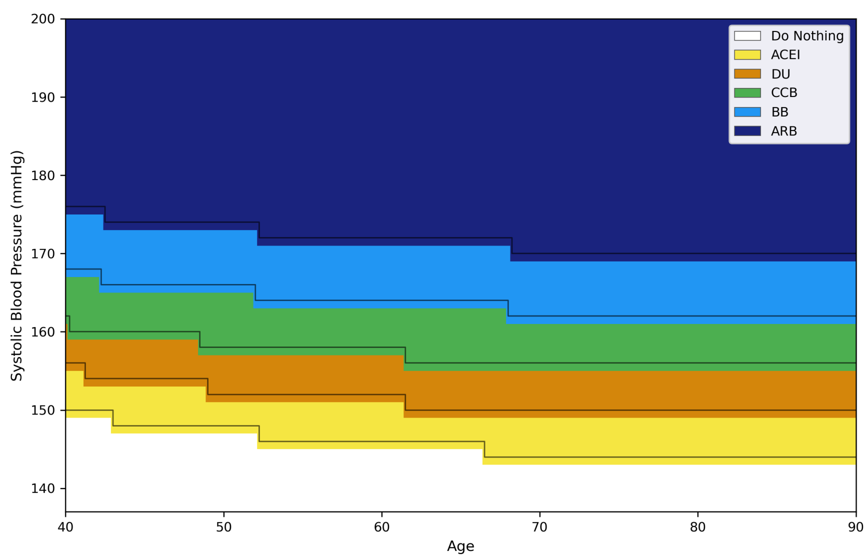


Figure 1 **Optimal medication prescription for a risk-free male under no arterial stiffness information (NI).**

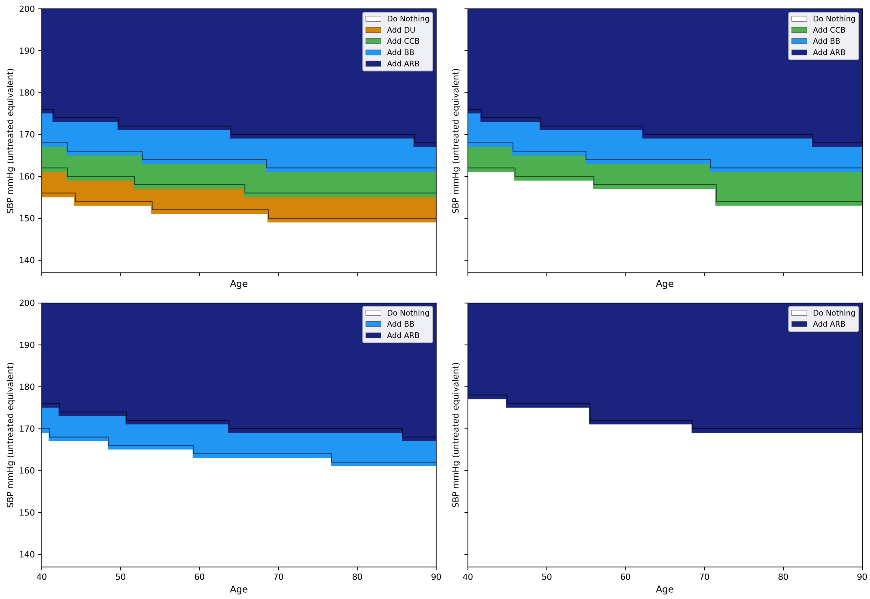


Figure 2 **Optimal medication prescription for a risk-free male under no arterial stiffness information (NI).**

Figure 3 incorporates the age-based stiffness proxy $w_{\text{ref}}(\text{age})$ used in decision-making where at 4 it is for prescribing medication by current regimen state of the patient.

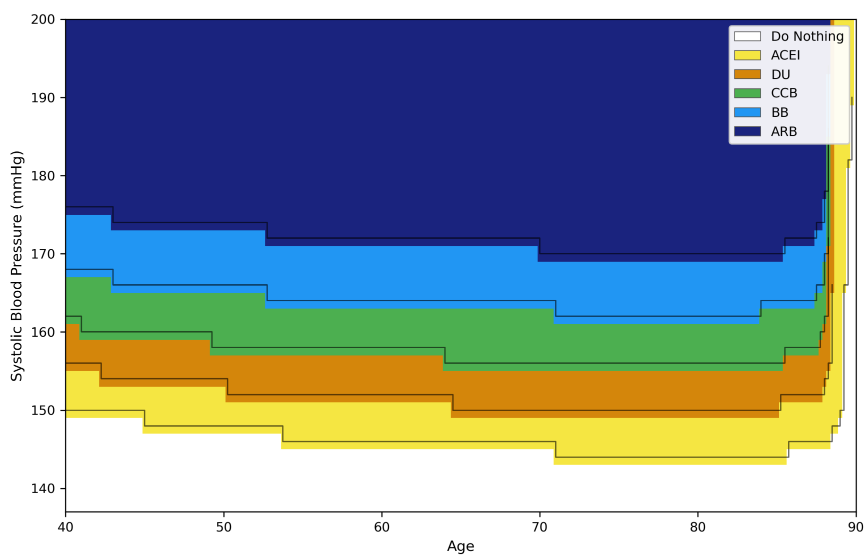


Figure 3 **Optimal medication prescription for a risk-free male under Preference-PWV proxy (PI).**

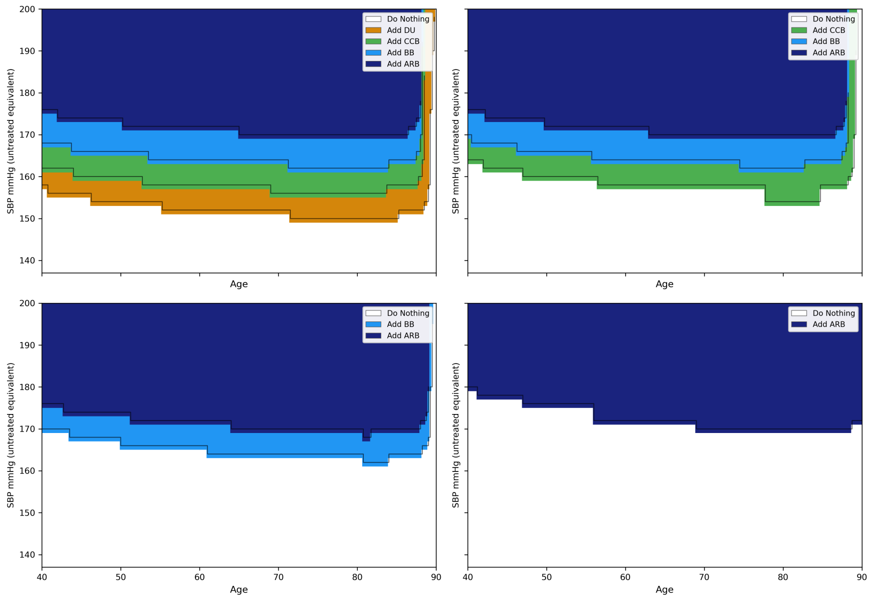
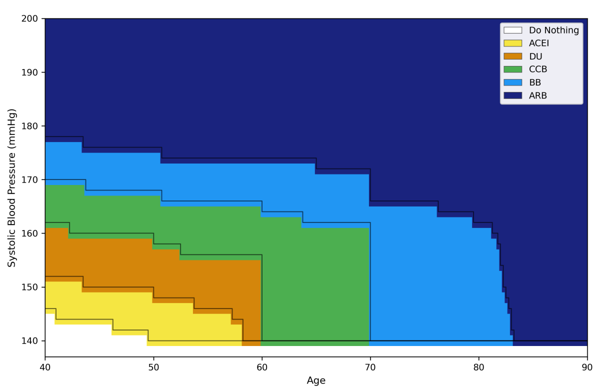
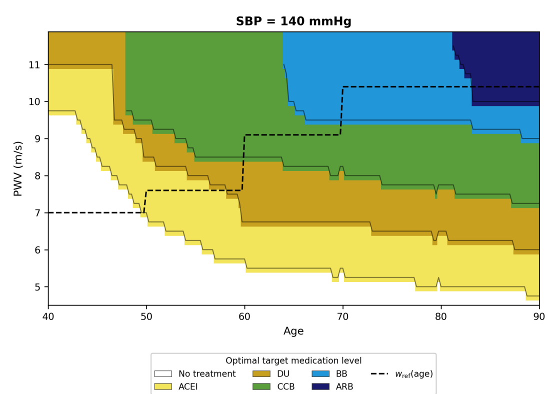


Figure 4 **Optimal medication prescription for a risk-free male under Preference-PWV proxy (PI).**

Figure 5 shows the policy when the true PWV state is observed (FI). Because the full policy depends on (b, m, w), causing a three-dimensional (SBP, medication, PWV) state space, the visualization is shown using two slices: one along the reference vascular aging trajectory $w = w_{\text{ref}}(\text{age})$ and one at a fixed SBP level. We display the optimal action as a function of age and PWV.



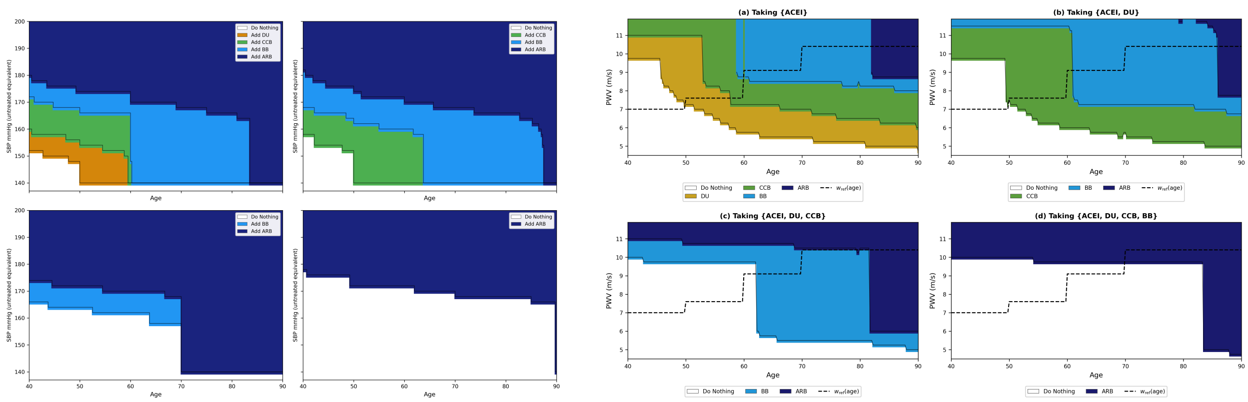
(a) Slice at $w = w_{\text{ref}}(\text{age})$



(b) Slice at SBP = 140

Figure 5 **Optimal medication prescription for a risk-free male under observed PWV dynamics (FI).**

Prescribing medication by current regimen state of the patient for the full information is based on PWV and SBP slices is shown below. More SBP slices are available upon request.



(a) Slice at $w = w_{\text{ref}}(\text{age})$ (b) Slice at SBP = 140

Figure 6 **Optimal medication prescription for a risk-free male under observed PWV dynamics (FI), conditional on current treatment regimen.**

Relative to the latent-PWV benchmark, incorporating stiffness information shifts treatment thresholds slightly downward at older ages, reflecting the increase in cardiovascular risk associated with vascular aging.

4.3. Policy Comparison Across Risk Profiles

Smoking male. Smoking substantially increases predicted cardiovascular risk, which lowers treatment thresholds across all PWV models.

Figure 7 displays the optimal initiation policy when PWV is ignored.

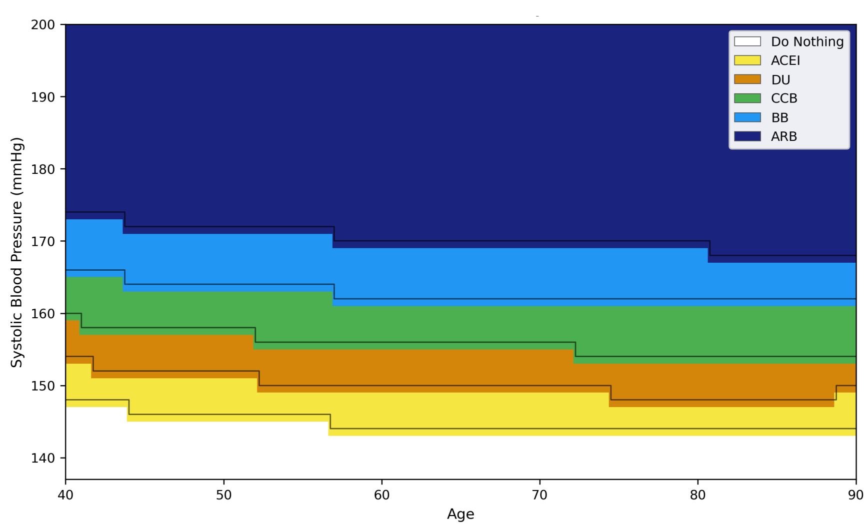


Figure 7 **Optimal first-line initiation policy for a smoking male under Policy A (latent PWV).**

Figure 8 shows the corresponding policy under the reference-PWV proxy model.

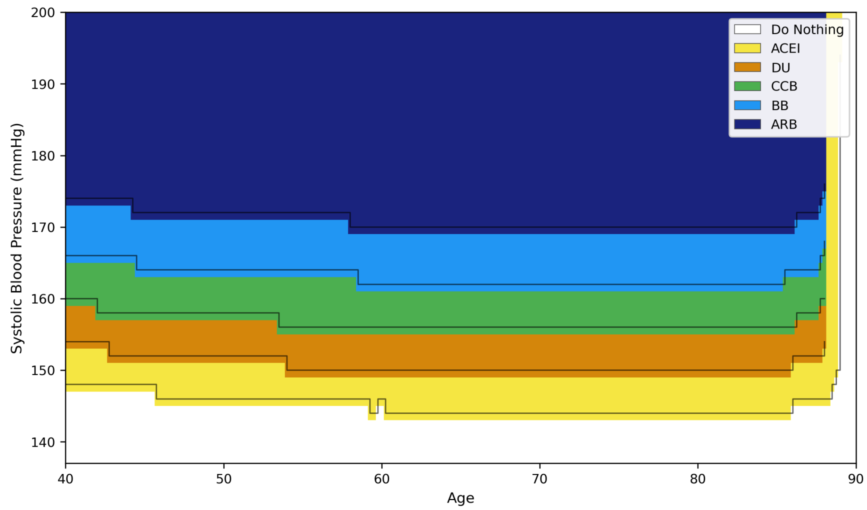


Figure 8 **Optimal first-line initiation policy for a smoking male under Policy B (reference-PWV proxy).**

Finally, Figure 9 displays the policy when PWV is fully observed.

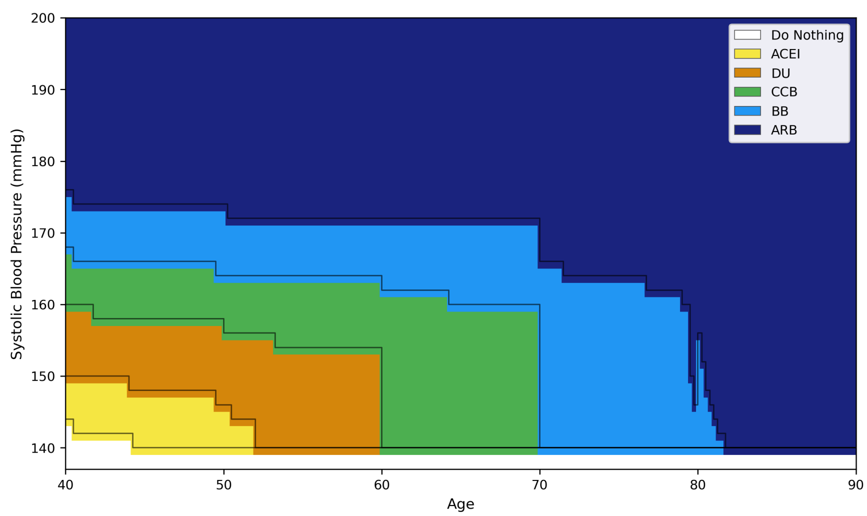


Figure 9 **Optimal first-line initiation policy for a smoking male under Policy C (observed PWV).**

Across all PWV models, the *Do Nothing* region shrinks substantially relative to the risk-free profile, indicating earlier treatment initiation at lower SBP levels.

Diabetic male. Diabetes also increases cardiovascular risk, though the magnitude of the threshold shift is generally smaller than that observed under smoking.

Figure 10 shows the initiation policy when PWV is ignored.

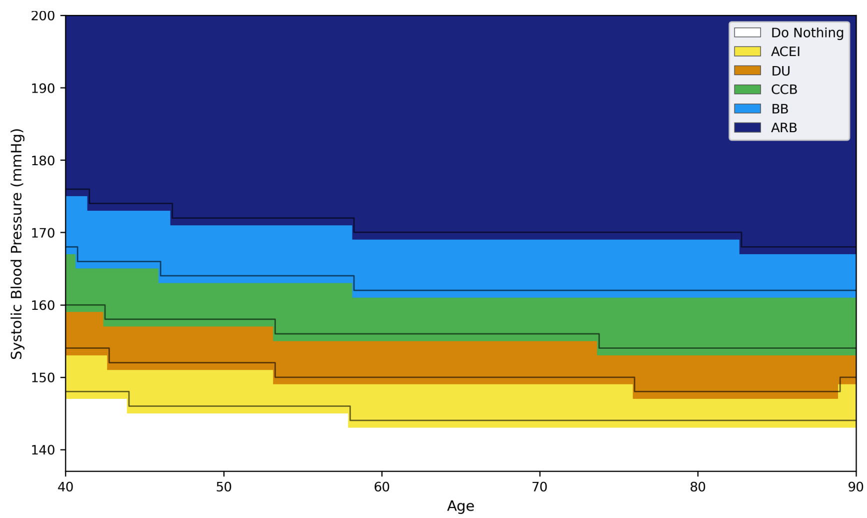


Figure 10 Optimal first-line initiation policy for a diabetic male under Policy A (latent PWV).

Figure 11 incorporates the age-based stiffness proxy used in Policy B.

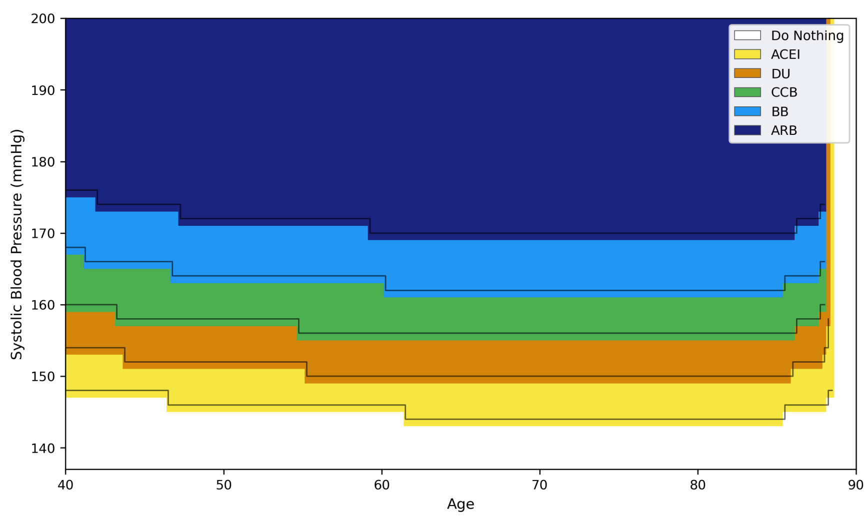


Figure 11 Optimal first-line initiation policy for a diabetic male under Policy B (reference-PWV proxy).

Figure 12 shows the fully observed PWV model.

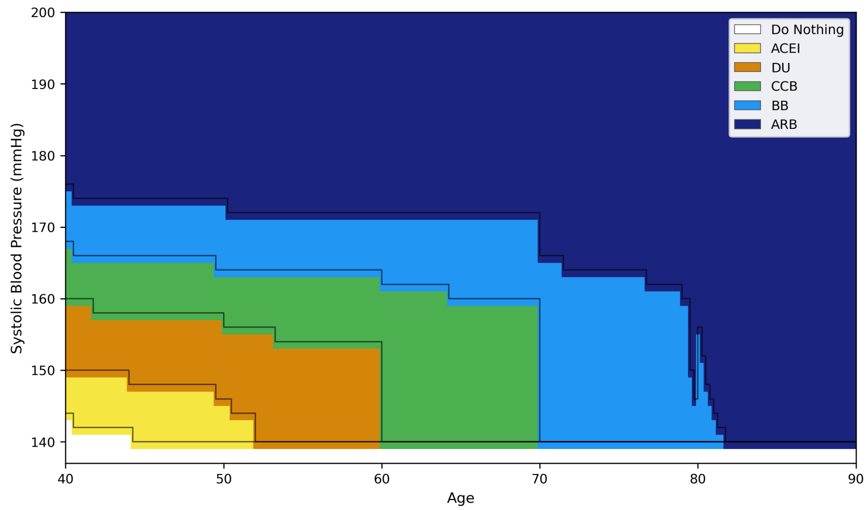


Figure 12 **Optimal first-line initiation policy for a diabetic male under Policy C (observed PWV).**

Relative to the risk-free profile, diabetes lowers initiation thresholds across most ages, though the shift is less pronounced than that induced by smoking in the calibrated risk model.

4.3.1. Dynamic PWV: SBP-Sliced Policy Visualisation In the dynamic-PWV specification, arterial stiffness w_n is an included patient state variable that evolves over time in a stochastic manner. Therefore, optimal policy is dependent upon the medication status, SBP, and PWV jointly. For any given medication state m and arterial stiffness w, there exists a threshold SBP above which optimal therapy escalation is to be provided.

As the state space consists of three dimensions (SBP, medication state, and PWV), we will use SBP slices to visualize the policy. Each SBP slice fixes the SBP level and illustrates the optimal action based on the subject's age (horizontal axis) and arterial stiffness (vertical axis). The colour of the panels represents the level of optimal medication; the dashed curve represents the reference stiffness trajectory $w_{\text{ref}}(\text{AGE})$.

Risk-free male.

At the lowest SBP level (140 mmHg), treatment is indicated only for patients who are both older and have an elevated PWV. The lower left portion of the diagram will, therefore, have a predominance of *Do Nothing* therapy for younger patients with normal arterial stiffness, who are at relatively low cardiovascular risk.

When arterial stiffness increases along the vertical axis of the SBP slices, the treatment threshold moves to the left within the SBP slices (i.e., treatment is initiated earlier for patients with higher

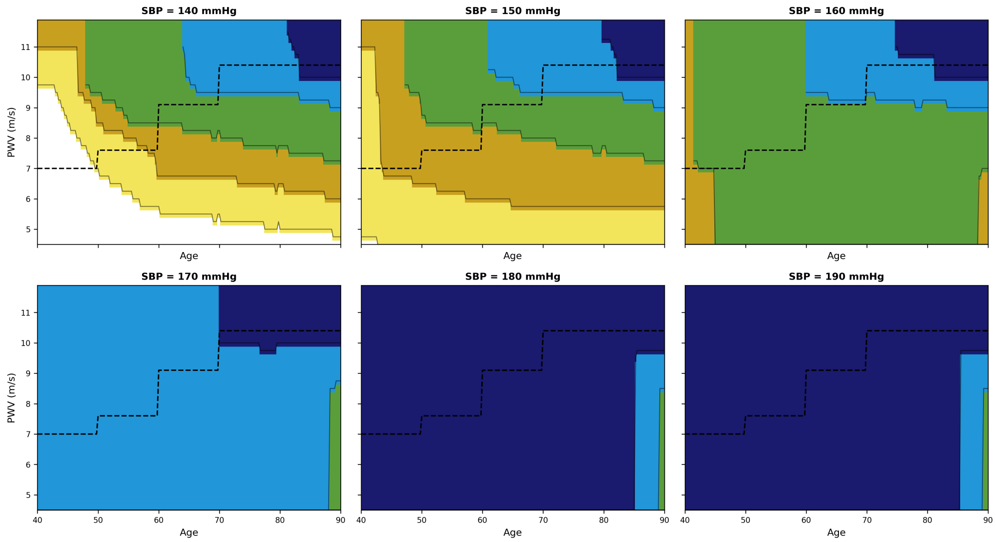


Figure 13 **Dynamic-PWV first-line initiation — risk-free male. Each panel fixes SBP and shows the optimal action as a function of age (x-axis) and PWV (y-axis).**

arterial stiffness). This shows that arterial stiffness independently shifts the treatment initiation threshold, such that patients with higher PWV will enter the treatment region earlier at a fixed SBP.

The range of indicated treatment expands along the total diagram as Systolic Blood Pressure (SBP) increases to 150-160 mmHg. A progressively flatter line delineating areas of indicated treatment from areas of no indicated treatment indicates that SBP will become the leading factor in making treatment decisions at 170 mmHg and greater. As SBP approaches 180-190 mmHg, the vast majority of the ages and Pulse Wave Velocity (PWV) will be considered indicated for treatment using existing therapies. This overall expansion supports the conclusion that SBP will be the primary decision-making tool in prescribing medication for patients, while PWV will provide fine-tuning of the decision point at the lower intermediate SBP level.

Figure 14 illustrates the escalation structure when SBP is fixed at 160 mmHg and patients are already receiving treatment. Each panel corresponds to a different medication state.

When an ACEI is the only drug regimen being used by the patient, the model typically indicates an indication to start a Diuretic or a Calcium Channel Blocker for the next step in therapy when escalated. Treatment escalation is first initiated among patients with greater PWV or older age, again demonstrating the degree of influence that increased stiffness has on the rate of medication intensification.

Medication burden increases as patients go through their first and second supplemental drug regimens $(\text{ACEI} + \text{DU})$, $(\text{ACEI} + \text{DU} + \text{CCB})$. Regions of indicated medication regimen escalation

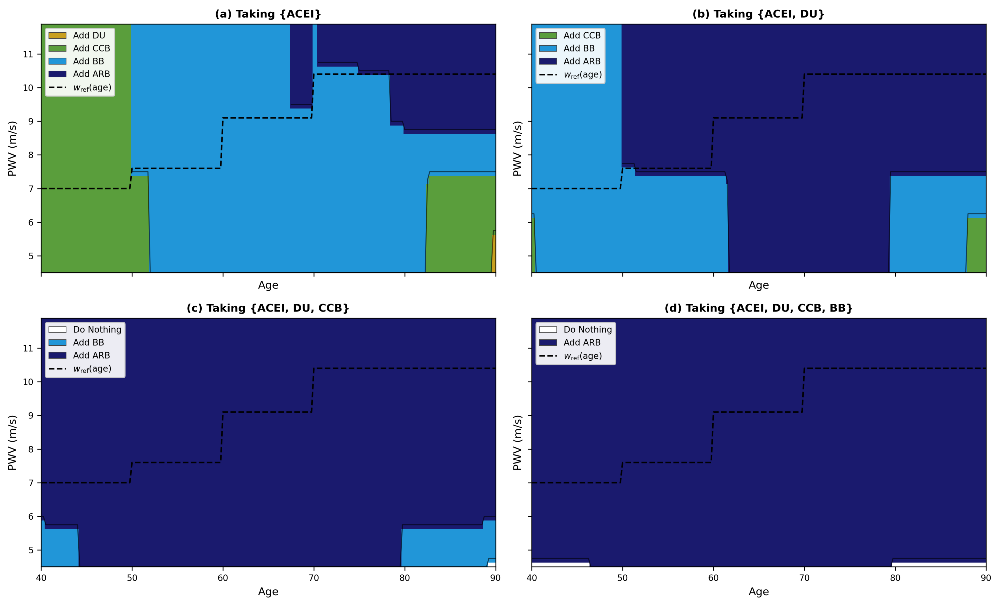


Figure 14 **Dynamic-PWV escalation at SBP = 160 mmHg — risk-free male, by current regimen state.**

then shrink significantly after three-drug regimens are reached. When a patient is using three medications, continuing on that regimen is frequently the best recommendation unless they have both a high PWV and are old in age at that time. When patients reach a four-drug medication regimen, the recommendation is nearly always to continue with the existing treatment since marginal improvement will result from the addition of any other medication(s).

Risk-free female.

The female SBP slices indicate generally larger *Do Nothing* zones than for males for the majority of the panels examined. For example, for SBP levels of 140 mmHg and 150 mmHg, treatment is generally only recommended for patients with much higher PWV and older age.

The same PWV gradient exists in each panel. As SBP is held constant, the treatment boundary descends with increasing age, where a higher PWV causes the treatment initiation decision to occur for younger patients. This independent contribution to the risk estimation of PWV demonstrates that the strength of the vessel wall contributes to risk, even if the baseline risk overall is lower.

At SBP levels of 170 mmHg or more, the vast majority of the state space will represent optimal treatment, which mimics the structure of male treatment policy but only occurs at slightly higher SBP levels.

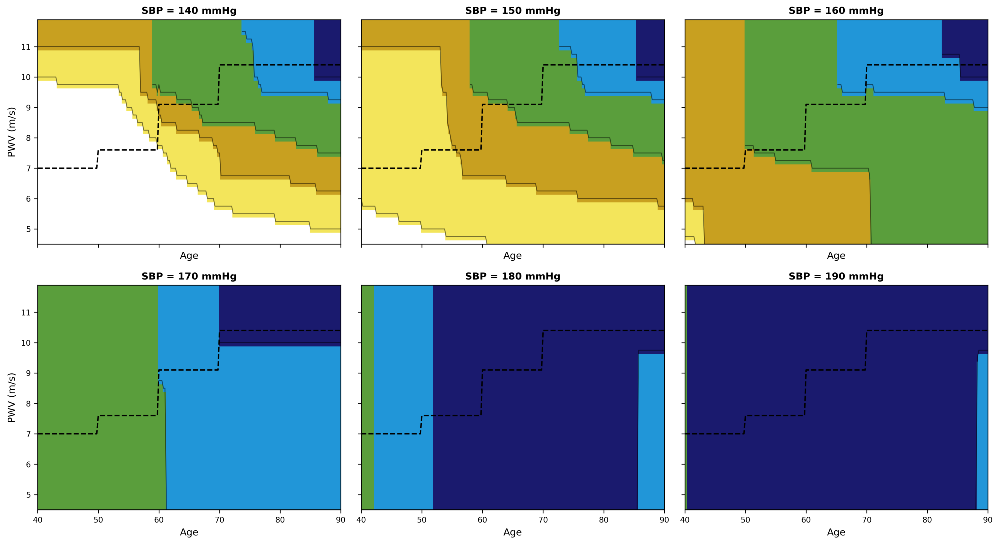


Figure 15 **Dynamic-PWV first-line initiation — risk-free female.**

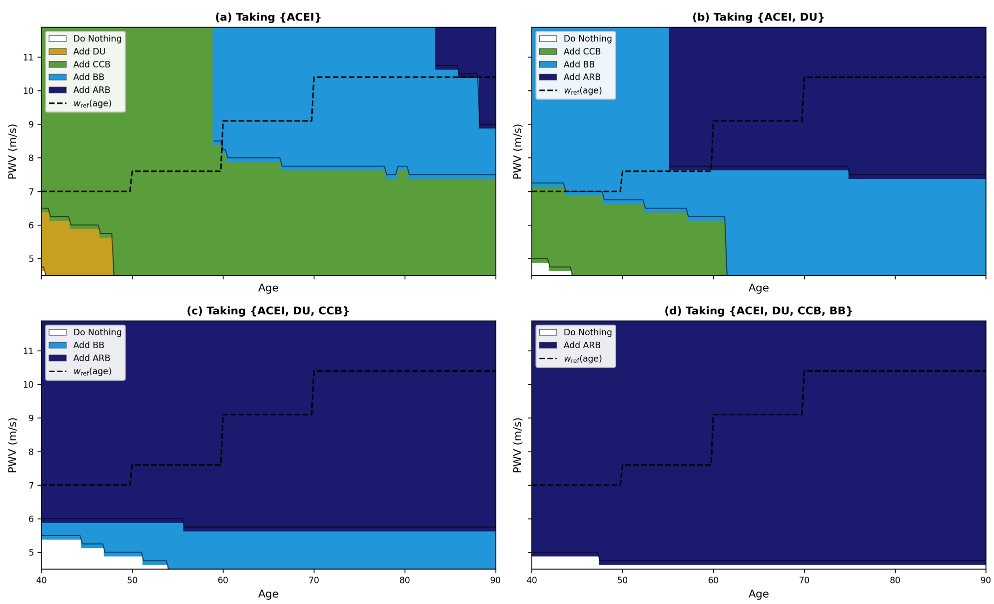


Figure 16 **Dynamic-PWV escalation at SBP = 160 mmHg — risk-free female.**

The female SBP escalation patterns are similar to those of males, although they occur in a smaller region of the state space. Patients with low PWV generally do not escalate their treatment, while escalation primarily occurs for older patients with much higher vascular stiffness.

Smoking male and female.

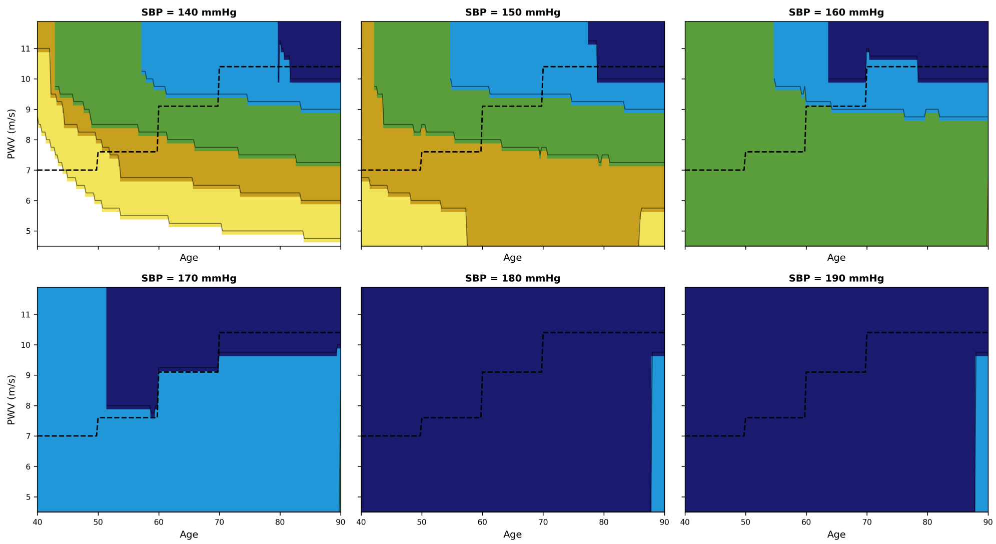


Figure 17 **Dynamic-PWV first-line initiation — smoking male.**

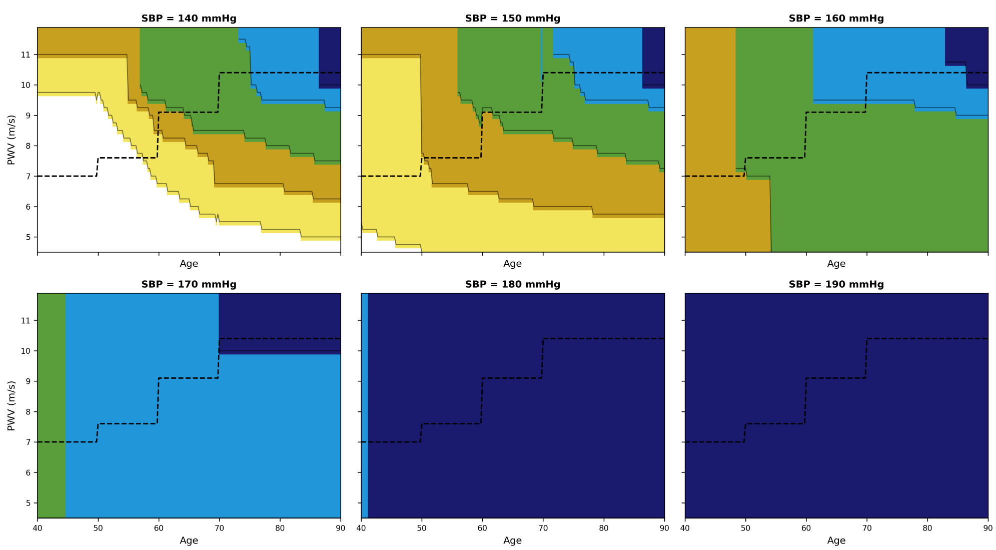


Figure 18 **Dynamic-PWV first-line initiation — smoking female.**

For patients who smoke, treatment regions expand dramatically across the entire SBP range. Even at SBP levels near 140 mmHg, treatment is optimal for a large portion of the age-PWV space.

The relationship between smoking and PWV can be seen most clearly in the vertical (Y-axis) direction. When the PWV of a smoker is elevated, their treatment boundaries shift downwards from that of a non-smoker.

Diabetic male and female.

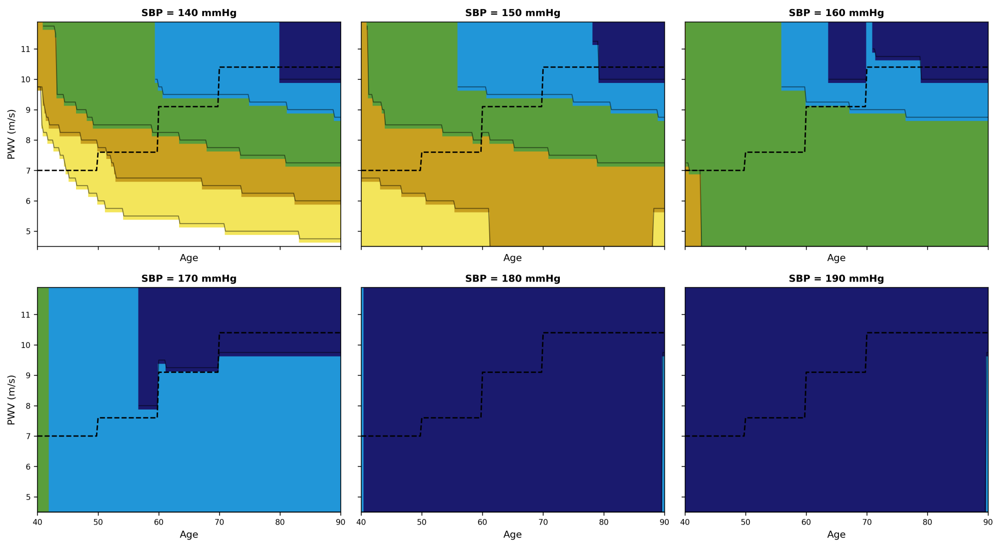


Figure 19 Dynamic-PWV first-line initiation — diabetic male.

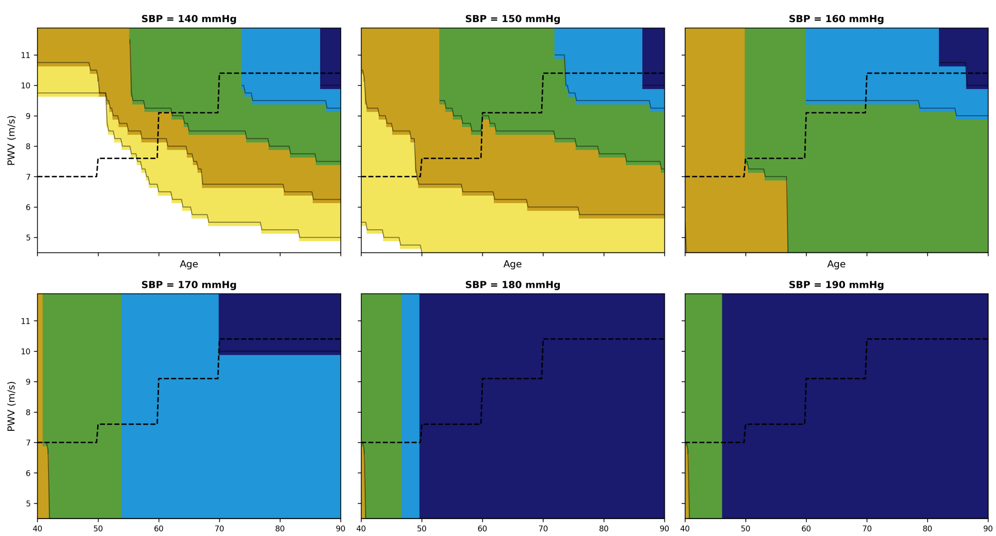


Figure 20 Dynamic-PWV first-line initiation — diabetic female.

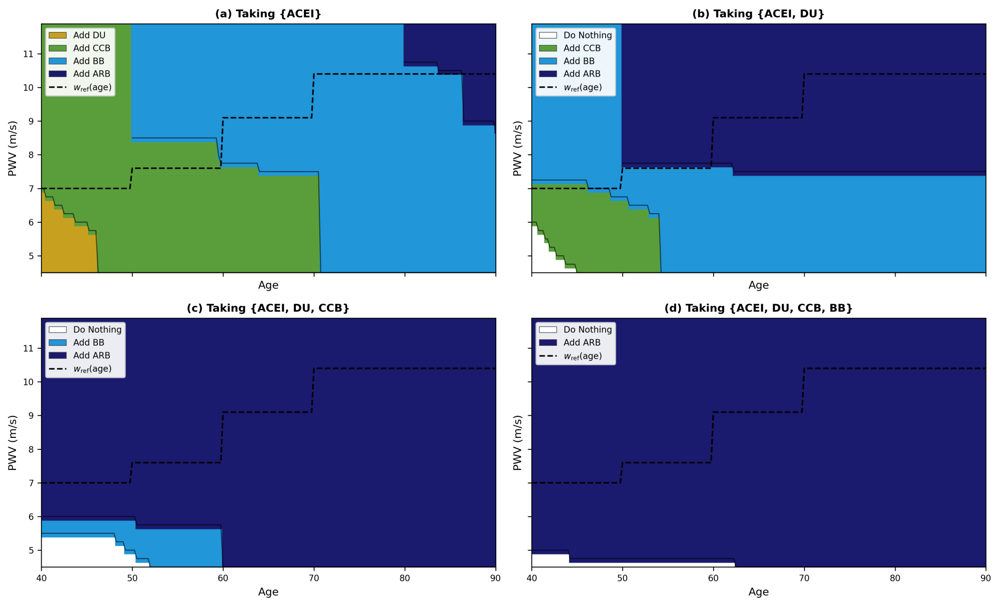


Figure 21 **Dynamic-PWV escalation at SBP = 160 mmHg — diabetic female.**

The earliest the boundaries for treatment initiation occur for smokers are when they have elevated PWV and they are older.

The diagnoses of patients with diabetes show an intermediate level of PWV in that they begin their treatment earlier than that of a non-smoker but later than a smoker at the same PWV/SBP level.

The effects of PWV on initiation of treatment are visible regardless of established levels of SBP. Higher PWV causes the initiation of treatment for younger patients and higher SBP levels than in a patient with no evidence of hypertension.

Summary of policy structure observations.

— The optimal policy retains a threshold structure in SBP across all patient profiles and both PWV specifications. The primary determinant of whether or not a patient will receive treatment remains the SBP.

— Under static-PWV, higher-risk profiles (smoking > diabetic > risk-free) shift thresholds downward, while female profiles shift thresholds upward relative to males. In particular, diabetic women have been shown to initiate treatment earlier statically compared to women without evidence of diabetes, but they still require a longer time before initiating treatment compared to patients who smoke.

— Under dynamic-PWV, PWV constitutes an *independent* second threshold dimension. At any fixed SBP, higher PWV shifts the policy toward earlier initiation and lower escalation thresholds. The addition of PWV therefore provides an additional mechanism for determining the point at which treatment should begin.

— A profile with a higher incidence of risk factors results in a movement of the treatment threshold down the SBP axis, while elevated PWV moves the treatment threshold to a younger age. Thus, patients with higher PWV or higher risk-factor burden will enter the treatment region earlier than lower-risk individuals.

— Escalation thresholds rise monotonically with medication burden, reflecting diminishing marginal returns (Timbie et al. 2010).

5. The Value of PWV Information

We compare three information structures for cardiovascular risk assessment and treatment evaluation:

— **Model A**: PWV ignored

— **Model B**: PWV approximated using age-based reference values

— **Model C**: observed PWV dynamically incorporated

All three models share the same treatment policy and the same SBP evolution dynamics. Differences therefore isolate the decision value of *observing* arterial stiffness, rather than inferring it coarsely from age or omitting it altogether.

The cohort contains $N = 84$ patients from the Montreal General Hospital dataset.

5.1. Cohort Analysis Framework

In this section we evaluate the operational value of arterial stiffness observability. Using data from the Montreal General Hospital dataset, we conduct a cohort-level analysis of how different information structures affect risk assessment and treatment evaluation when the underlying clinical decision rule is fixed. The goal is not to change the treatment policy itself.

We simulate treatment recommendations and predicted outcomes for each patient using three different models: The first one ignores PWV (Model A), the second approximates PWV based on age-based reference values only (Model B), and the third one has fully observed PWV (Model C). As the treatment policy and SBP evolution dynamics are fixed across all models, the resulting differences in predicted risk/treatment classification/clinical outcomes are due solely to the difference in information structures.

This comparison isolates the decision value of observing arterial stiffness. The analysis evaluates whether additional information from measuring PWV has improved risk stratification/treatment allocation beyond what can be inferred by age and blood pressure alone in an operational context.

To characterize this value, we look at many outcomes, such as predicted cardiovascular risk; number of events prevented by treating (treatment status classification as ”underserved” or match or ”over-serve”) and differences between patient treatment across Models. There are thus a variety of measures by which to evaluate the overall value of arterial stiffness observability.

5.2. Empirical Consequences of PWV Observability

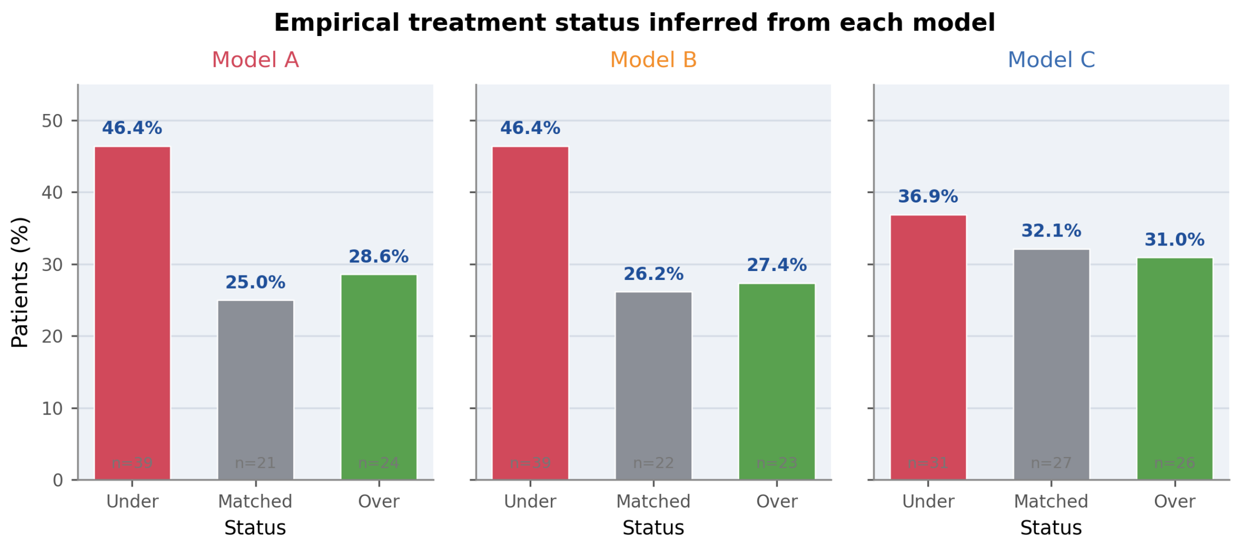


Figure 22 **Empirical treatment status inferred from each model.**

5.2.1. Observed PWV changes treatment classification We begin with the most operationally important question: how does the information structure change the model’s assessment of whether current treatment is adequate?

Figure 22 classifies each patient as *undertreated*, *matched*, or *overtreated* relative to the treatment intensity implied by the model. Models A and B produce nearly identical treatment-status profiles. Both classify 46.4% of patients as undertreated, while only 25.0% (Model A) and 26.2% (Model B) are judged appropriately matched.

Once observed PWV is incorporated, the classification changes materially. Under Model C, the undertreated share falls from 46.4% to 36.9%, while the matched share rises to 32.1%. The overtreated share also rises modestly to 31.0%, reflecting that observed stiffness reallocates some patients away from the coarse treatment assignments implied by SBP and age alone.

The managerial implications of learning vascular stiffness based on the observed state are obvious; if they aren't present or observable, we will assess the adequacy of our treatment based only on incomplete data. Models based solely on age or systolic blood pressure tend to overestimate the degree to which a patient is not treated, and miss providing us with differentiated information about which patients represent the greatest risk. Observation increases not just the ability to predict risk of an event, but also the logic for determining which patients should be flagged for treatment intensification.

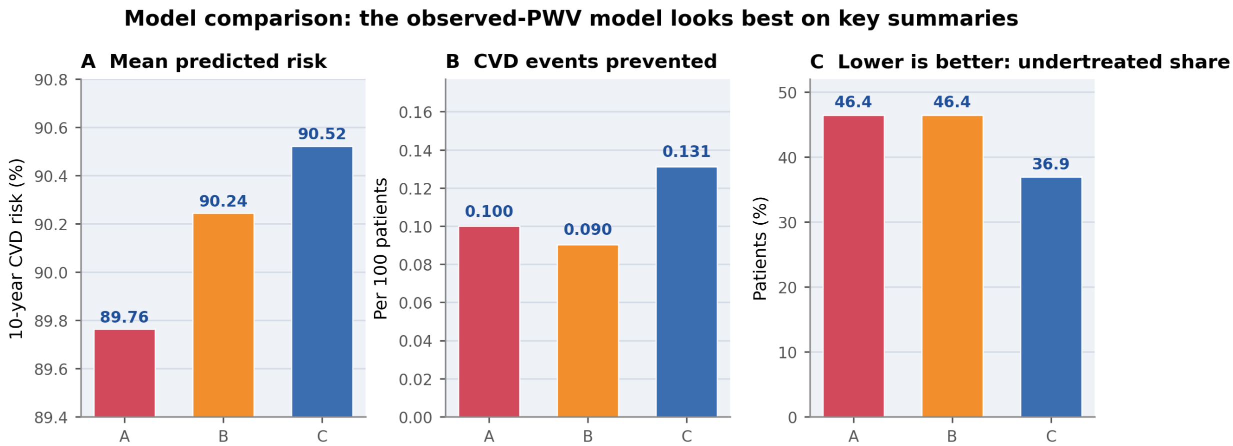


Figure 23 **Comparison of key performance metrics across models.**

5.2.2. Observed PWV improves aggregate model performance Figure 23 summarizes the aggregate consequences of moving from latent or proxy PWV to fully observed PWV. Model C performs best on all three summary metrics.

First, Model C produces the highest mean predicted 10-year CVD risk (90.52%), compared with 90.24% for Model B and 89.76% for Model A. Second, it yields the largest number of prevented cardiovascular events (0.131 per 100 patients), exceeding both Model A (0.100) and Model B (0.090). Third, it produces the lowest undertreated share (36.9%, versus 46.4% under both Models A and B).

Importantly, the treatment policy itself is held constant across models. The improvement therefore arises from better *state measurement*, not from a more aggressive policy. In operational terms, measured PWV acts as a higher-quality signal that improves prioritization, targeting, and treatment adequacy assessment.

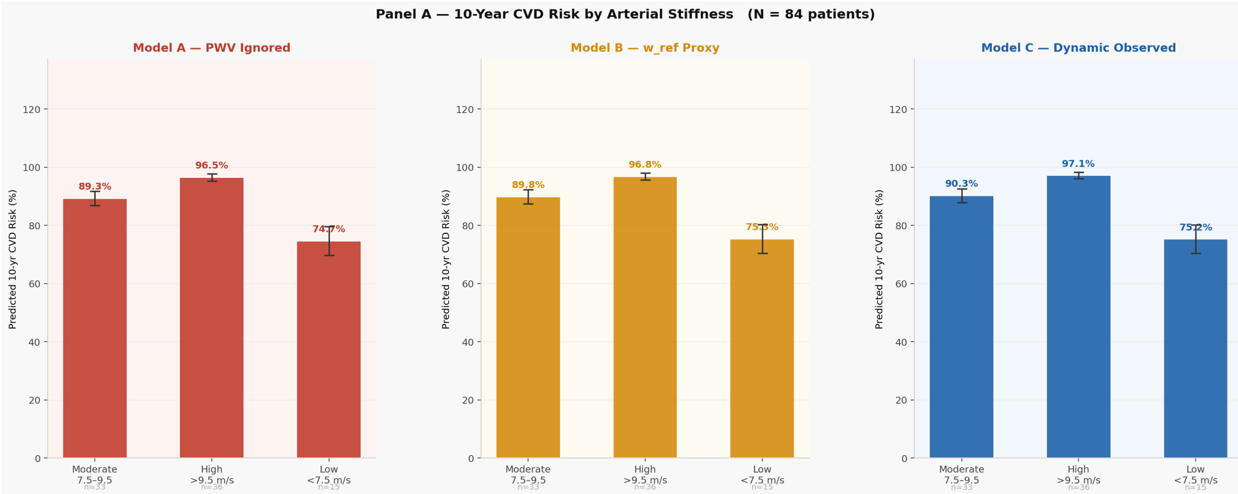


Figure 24 **Predicted CVD risk by arterial stiffness group.**

5.2.3. Observed PWV reveals biological heterogeneity beyond age Figure 24 illustrates the mechanism behind these improvements. Predicted cardiovascular risk rises sharply across arterial stiffness groups. Patients in the high-stiffness group (PWV > 9.5 m/s) exhibit substantially higher predicted risk than those in the low-stiffness group (PWV < 7.5 m/s).

The pattern shown demonstrates that PWV encompasses clinically important variations that potential age alone cannot represent. The risk position of patients who are the same age and have the same SBP can vary significantly once vascular stiffness is taken into account. Consequently, risk categories that do not utilize PWV may group together biologically dissimilar patients into risk categories not based on unique patient characteristics.

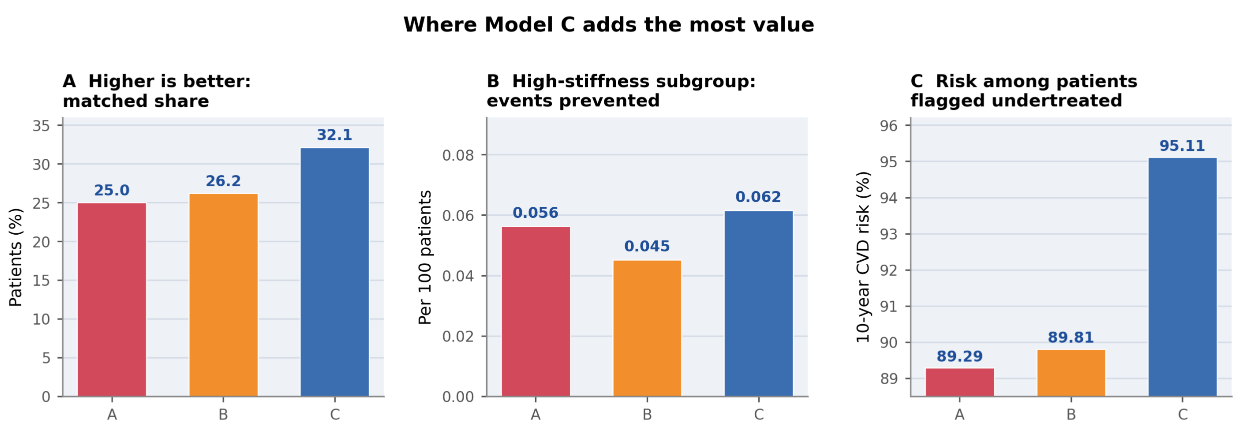


Figure 25 **Areas where the observed-PWV model provides the largest improvements.**

5.2.4. Where does PWV observability add the most value? Figure 25 identifies where Model C extracts the greatest benefit from improved observability. The observed-PWV model achieves the highest matched-treatment share (32.1%), the largest number of events prevented in the high-stiffness subgroup (0.062 per 100 patients), and the highest mean risk among patients flagged as undertreated (95.11%).

Model C can target the appropriate patients for treatment and allows for the treatment of those patients who actually have an increased risk for cardiovascular disease based on their biological vascular health.

The main managerial insight from the cohort analysis is that while all patients do not get treated differently based on their observability score alone, PWV will improve the quality of the treatment decision for patients whose biological vascular age is substantially different from their chronological age and for borderline patients where the decision is influenced the most by latitude due to damage not visible in other models.

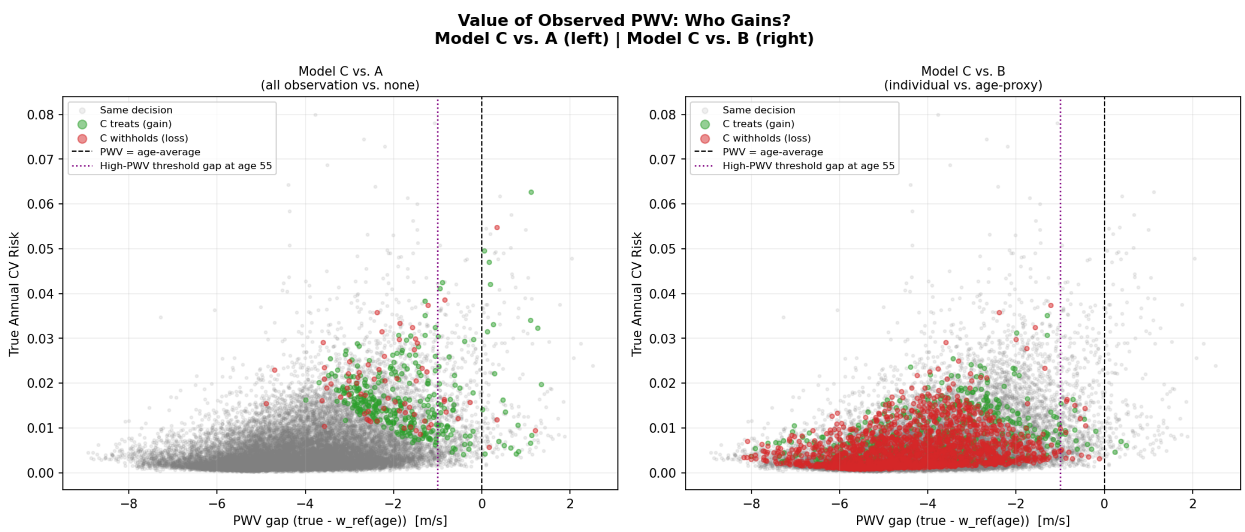


Figure 26 **Patient-period-level treatment differences between the observed-PWV model and simpler alternatives.**

5.2.5. Patient-level evidence: observed PWV separates biological from chronological age Figure 26 provides a patient-level view of the value of observability. The horizontal axis measures the PWV gap,

$$\Delta w = w_{\text{true}} - w_{\text{ref}}(\text{age}),$$

which captures how much a patient's realized arterial stiffness deviates from the age-based reference level.

When Model C is compared to Model A: most treatment differences are found to be in the top half of untreated patients whose PWV value exceeds the threshold expected at their age. Patients in this group exhibit biologically 'stiffer' characteristics than their physical age alone would imply. Thus, if there is an observed dominant treatment effect, there likely also would be a treatment effect with respect to Model B, where the majority of treated patients are documented within the negative gap area (i.e., treated according to age-based proxy but not treated according to observed-PWV model). Patients in this group have essentially received overtreatment (i.e., too much treatment) based on the use of the proxy for how stiff they may look based on their age.

Because an observed PWV is critical for the decision-making process (especially for determining who is 'stiff'), observed PWV is also the best method of differentiating among true 'stiff' patients and those who have age proxies, thereby allowing for optimal targeting of treatment resources and reducing the risk of under- or over-treating patients.

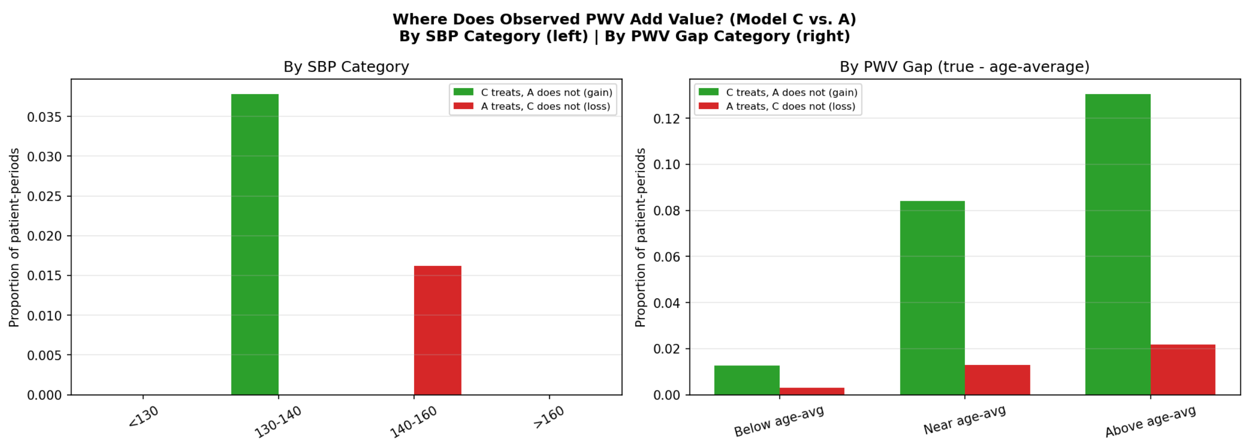


Figure 27 **Subgroup decomposition of where observed PWV changes treatment decisions.**

5.2.6. Subgroup decomposition of gains from observability Figure 27 shows that the gains from PWV observability are concentrated in a narrow but clinically important region of the state space.

The left-hand chart of the figure illustrates the divided agreement by systolic blood pressure (SBP) range. The vast majority of Model C's advantages relative to Model A come from patients between 130 and 140 mmHg SBP. For SBPs less than 130 mmHg, neither model provides treatment;

for those above 160 mmHg, both models are highly likely to provide treatment. Therefore, above the intermediate SBP (neurological) range, PWV becomes most relevant only to patients at the border of treatment determined by SBP.

The chart on the right shows the benefits from Model C versus Model A separately for individuals depending on their difference between their applicable and actual PWV. The greatest improvements to Model C over Model A are occurring in the "greater than applicable (age average) PWV" category. This demonstrates that the usefulness of using observable evidence through the application of both PWV and SBP lies primarily with patients have a SBP that is close to the threshold of being treated, and that have an actual vascular stiffness that exceeds their expected (age-predicted) vascular stiffness.

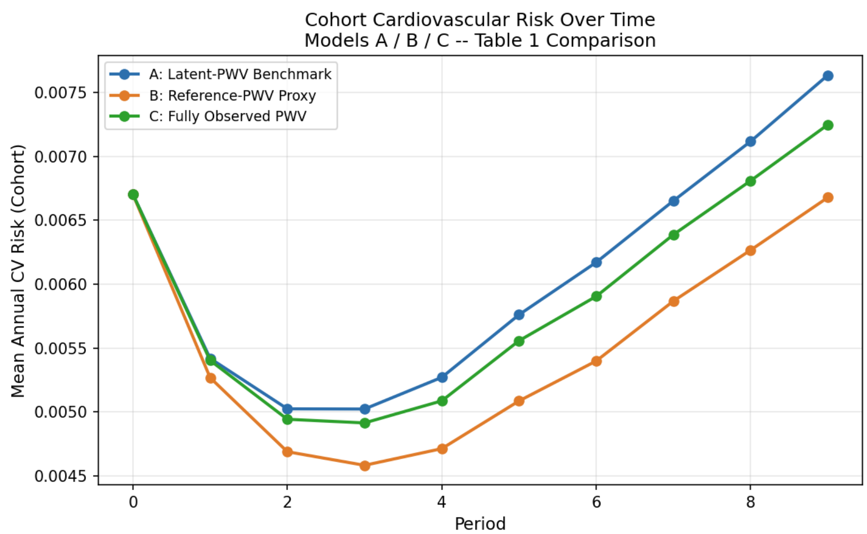


Figure 28 **Mean cohort cardiovascular risk over time under Models A, B, and C.**

5.2.7. Simulation evidence over time Figure 28 tracks mean annual cardiovascular risk over the simulated cohort across decision periods. All three models begin at the same baseline risk and then diverge as treatment choices alter SBP trajectories over time.

The early decline in blood pressure due to treatment shows how effective this is initially while the increase later may be due to age and the gradual rise in stiffness of arteries as they age. Model B has the lowest overall risk of all cohorts over time, as it is able to apply broader treatment than models A and C because of the use of a greater degree of age-homogeneity rule. Model C offers an overall cohort benefit but at a slightly lesser level than model B, offering the ability to target a narrower portion of older individuals based on their level of stiffness.

Model B does not contradict earlier cohort data because its broad application of a ”stiffness-adjusted” treatment approach results in a lower average risk level for the entire cohort, but its application to older cohorts includes some patients who have a lower than average level of severity. In contrast, model C has a higher degree of observability; therefore, the key advantage of observability is not to treat all patients more broadly but to treat patients with a higher level of accuracy and biological risk based on their observations.

5.3. QALY Value of PWV Information: Models NI, PI, and FI

As determined by the previous section regarding how many quality-adjusted life years (QALYs) would result from observed changes in arterial stiffness (regardless of what type of treatment was prescribed), we will now proceed to calculate the ’value’ of the arterial stiffness changes. To accomplish this, we will perform the complete backward-induction to solve the problem for each of the three information structures shown in Table 6, for six different types of risks (for our example patients), while maintaining a constant policy structure throughout.providing only information about the stiffness of the arteries as an indication of what actions (if any) should be taken by the decision maker after having made the ’decision’ that such information is available.

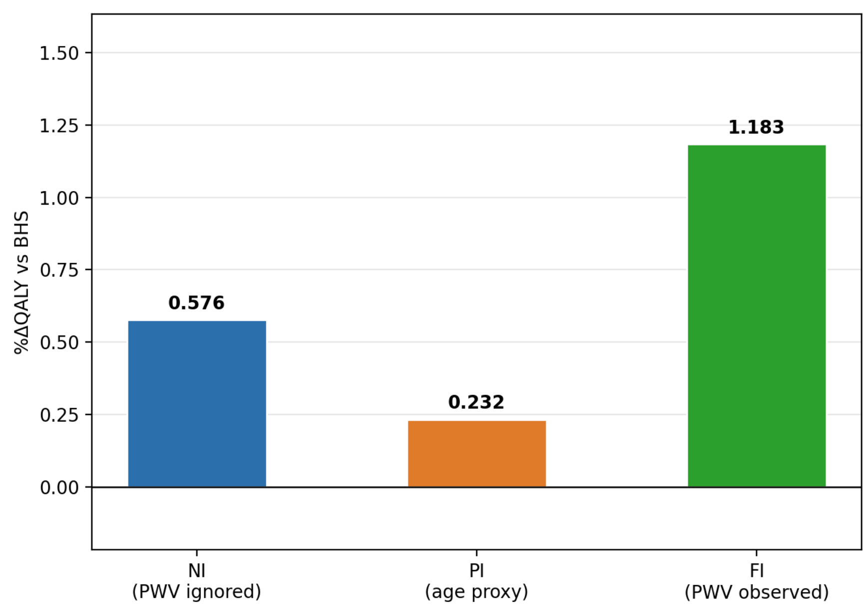


Figure 29 **Average Percentage QALY improvement of Models NI, PI, and FI relative to the BHS rule-based benchmark, averaged across the six patient risk profiles.**

Figure 29 summarises the average welfare gains of the three information structures relative to the BHS guideline. On average, Model FI (fully observed PWV) delivers a 1.183% improvement in expected QALYs, substantially exceeding the gains of Model NI (0.576%) and Model PI (0.232%). These results highlight two distinct sources of improvement relative to rule-based treatment. First, dynamic optimisation alone produces meaningful benefits: even without any stiffness information, Model NI improves expected QALYs by more than half a percent relative to BHS. Second, the availability of patient-specific arterial stiffness information generates an additional layer of value, more than doubling the average improvement when moving from NI to FI. By contrast, the age-based proxy (Model PI) captures only a small fraction of this informational value, indicating that coarse physiological proxies provide limited decision support relative to direct measurement.

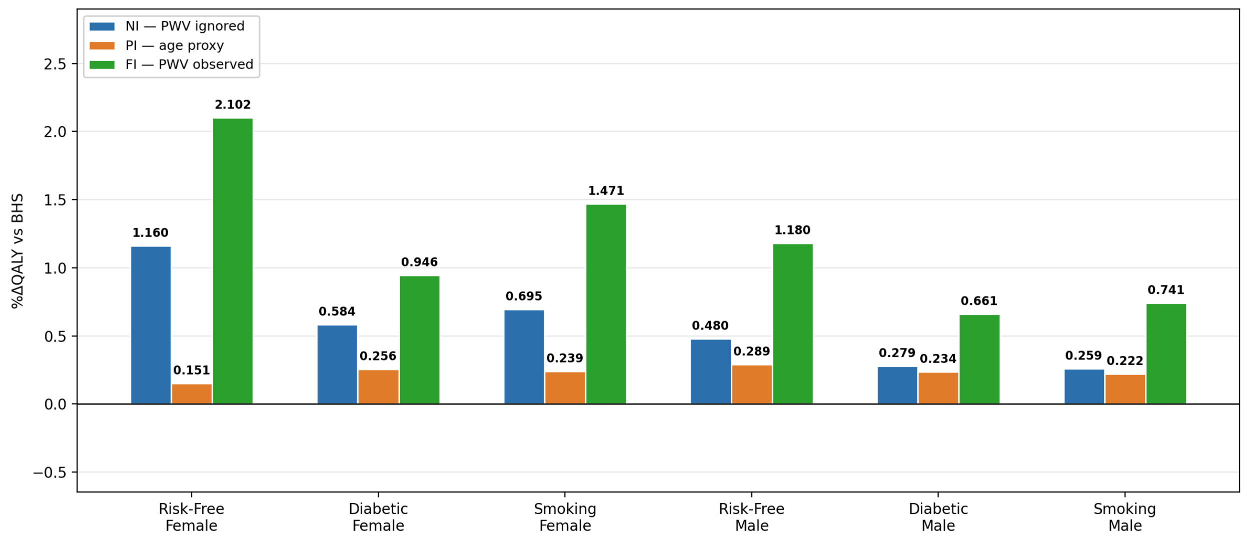


Figure 30 **Percentage QALY improvement of Models NI, PI, and FI relative to the BHS rule-based benchmark, for six patient risk profiles. All three models use flexible medication sequences with age in the state space; they differ only in the stiffness signal available to the decision maker.**

Figure 30 reports the welfare improvements for each individual risk profile. All three optimisation-based models outperform the BHS guideline across every profile, confirming that replacing static rule-based treatment with a dynamic optimisation framework yields systematic improvements in expected patient outcomes.

However, the magnitude of the gains differs substantially across the three information structures. Model FI consistently achieves the largest improvements in every profile, with gains ranging from 0.661% for the diabetic male to 2.102% for the risk-free female. The improvements are particularly pronounced in lower-risk populations, such as risk-free females and smoking females, where the FI model produces gains exceeding 2.1% and 1.47%, respectively. This pattern reflects a precision effect: when baseline cardiovascular risk is relatively low and treatment decisions are finely balanced, accurate physiological information becomes especially valuable for determining whether treatment escalation is warranted.

Model NI, which ignores arterial stiffness entirely, occupies a consistent intermediate position. Despite lacking any stiffness signal, it still delivers substantial improvements relative to BHS, ranging from 0.259% (smoking male) to 1.160% (risk-free female). These gains arise purely from the dynamic optimisation of treatment decisions, demonstrating that a large portion of the welfare improvement relative to guideline-based care originates from optimised decision-making rather than additional clinical information alone.

Model PI, which approximates stiffness using an age-based proxy, produces the smallest improvements in nearly all cases. The gains range from 0.151% for the risk-free female to 0.289% for the risk-free male. In five of the six profiles, Model PI performs worse than Model NI, indicating that the proxy introduces distortions that partially offset the benefits of optimisation. Because age captures only the population-average progression of vascular stiffness, the proxy fails to reflect substantial inter-individual variation in arterial ageing, leading to treatment decisions that may systematically misrepresent true cardiovascular risk.

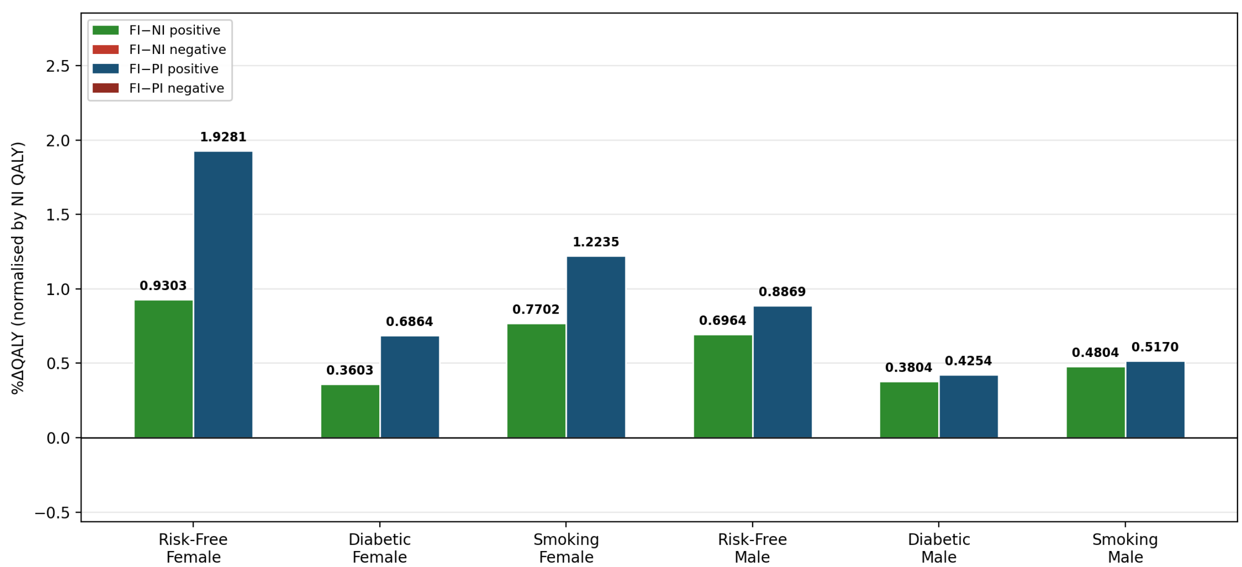


Figure 31 **Incremental value of observing individual PWV. The left bar shows FI−NI (the gain from observing PWV relative to ignoring stiffness), and the right bar shows FI−PI (the gain relative to the age-based proxy). Both quantities are normalised by the QALY of Model NI.**

Figure 31 decomposes the informational value of PWV measurement by comparing two incremental improvements: the gain from observing arterial stiffness relative to ignoring it (FI−NI), and the gain relative to the age-based proxy (FI−PI).

Across all six patient profiles, both improvement measures are strictly positive, indicating that direct observation of arterial stiffness improves expected outcomes relative to both ignoring stiffness information and relying on an age-based proxy. However, the magnitude of these improvements varies across patient profiles. The largest improvements occur for risk-free females, where the difference between FI and NI equals 0.930%, and the difference between FI and PI reaches 1.928%.

Substantial gains also appear for smoking females (FI−NI = 0.770%, FI−PI = 1.224%) and risk-free males (FI−NI = 0.696%, FI−PI = 0.887%). In contrast, smaller improvements are observed for higher-risk male profiles, particularly diabetic males, where FI−NI equals 0.380% and FI−PI equals 0.425%.

For diabetic male patients, treatment escalation is often warranted regardless of the availability of arterial stiffness information. Consequently, the marginal value of additional physiological signals is smaller in these high-risk populations.

A notable pattern also emerges across all six profiles. For every patient group, the improvement achieved by the fully observed model (FI) exceeds the improvements obtained under both the NI and PI models. In particular, the improvement of FI relative to PI consistently exceeds the improvement relative to NI. This result suggests that estimating arterial stiffness using age as a proxy can introduce larger welfare losses than ignoring stiffness information entirely, because the proxy embeds a noisy and potentially misleading estimate of vascular ageing into the decision process.

Three conclusions emerge from Figures 29–31.

The PWV-based model with full observation (Model FI) outperforms all alternative information structures across every patient profile. Relative to the BHS benchmark, Model FI produces QALY improvements ranging from 0.661% to 2.102%. These results indicate that incorporating patient-specific arterial stiffness information substantially improves the quality of treatment decisions.

Dynamic optimisation itself accounts for a large share of the welfare improvement. Model NI, which ignores arterial stiffness entirely, still achieves gains between 0.259% and 1.160% relative to guideline-based care. This finding suggests that much of the improvement arises from optimised treatment timing and sequencing within a dynamic decision framework rather than from additional physiological information alone.

In contrast, the age-based proxy for stiffness (Model PI) performs unreliably. For most patient profiles, Model PI performs worse than Model NI, indicating that age is an imperfect surrogate for arterial stiffness. Using age as a proxy introduces systematic distortions in the risk signal and can therefore lead to suboptimal treatment decisions.

Taken together, these findings highlight two complementary mechanisms that drive improvements in treatment policy design: optimised dynamic decision-making and accurate physiological measurement. Dynamic optimisation provides the baseline improvement relative to rule-based care, while incorporating patient-specific PWV information adds precision and further increases expected patient welfare. Because age captures only the population-average progression of vascular ageing, the proxy embeds a noisy signal into the decision process and can systematically misclassify patients whose true arterial stiffness deviates from the age-based expectation.

5.4. Why partial information is worse than full information?

The results in Figures 30–31 demonstrate that Model FI, which assumes full observation of arterial stiffness (PWV), consistently outperforms Model PI, which replaces individual PWV with an age-based proxy. The fundamental principle of uncertainty decision-making states that because each individual has a different underlying physiological signal, you cannot determine what the true state variable is by using a proxy.

The FI model has a major advantage in that the decision maker can see directly what the patient's stiffness is (w_t). Therefore, the treatment will be based directly on the vascular condition of each individual's stiffness. As arterial stiffness is an important determinant of cardiovascular risk, accurate observations can be used to identify patients whose risk will range from moderate to low due to their being "age-appropriate" but are still at risk from having a high blood pressure.

The treatment rule will optimally prescribe medications only to those patients for whom the expected benefit outweighs the disutility associated with the medication.

In contrast, the PI model substitutes a "deterministic" estimation of the stiffness state which is based upon chronic age ($w_{\text{ref}}(\text{age})$). While most patients become progressively stiffer with age, there is a large amount of variation in PWV values between patients of similar ages: i.e., due to genetics, lifestyle, co-morbidity, and/or cumulative damage to the arteries over time.

The PI model collapses this variability into a single deterministic value, so all patients within the same age cohort will have the same likelihood of developing the same medical condition independent of their risk.

The absence of valid PWV information leads to two distinct types of decision errors. First, patients whose true PWV is higher than the proxy estimate receive insufficient treatment because the approximation underestimates their cardiovascular risk. Second, patients whose true PWV is lower than the proxy estimate receive excessive treatment because the approximation overestimates their underlying risk. Both types of errors reduce overall health outcomes by shifting treatment decisions away from the optimal decision thresholds, thereby lowering expected QALYs.

The magnitude of this distortion is evident in the empirical results. Across all six patient profiles, the FI model generates larger QALY improvements relative to the BHS benchmark than the PI model. For example, for the risk-free female profile, the FI model achieves a 2.102% improvement relative to BHS, whereas the PI model produces only a 0.151% improvement. Similar patterns emerge across the remaining profiles. FI improvements range from 0.661% to 1.471%, while PI improvements never exceed 0.289%.

These differences highlight the welfare loss that arises when true patient-specific arterial stiffness is replaced with an age-based approximation. By relying on age as a proxy for PWV, the PI model introduces systematic misclassification of cardiovascular risk, which propagates through the treatment policy and ultimately reduces the achievable health gains relative to the full-information benchmark.

The breakdown of Figure 31 provides additional insight into the incremental value of directly observing PWV. The incremental gain from observing PWV directly—measured as the difference between the FI and NI models—ranges from 0.38% to 0.93% across patient profiles. In contrast, the incremental difference between the FI and PI models—where PWV is approximated using an age-based proxy—ranges from 0.425% to 1.928%. The substantially larger gap between FI and PI indicates that the proxy does not merely fail to recover the information contained in the true PWV state but can also introduce systematic distortions into treatment decisions.

From an operational perspective, this result reflects a classic *value-of-information* principle. In the FI model, PWV enters the state space directly and therefore influences the transition probabilities governing disease progression and cardiovascular risk. By contrast, the PI model relies on a proxy signal that is only imperfectly correlated with the true physiological state. When the signal-to-noise ratio of this proxy is low, incorporating it into treatment decisions may degrade decision quality. In such cases, the resulting policy can perform worse than a policy that relies exclusively on observable clinical variables such as systolic blood pressure and medication history.

The key takeaway from this data set is that although age is still important for assessing hypertension, it is not enough by itself to assess vascular health. When accurately measuring arterial stiffness, this provides clinically relevant value beyond simply using demographic data.

When PWV is used directly to guide treatment, there will be consistent improvement in treatment outcomes compared to using an indirect approximation based on demographic characteristics.

In summary, PWV not only assists in evaluating risk for hypertension but also helps improve the quality of decisions made when managing patients with hypertension. By resolving the variability in vascular stiffness at the patient level, the FI model allows for improved precision in selecting appropriate treatment plans, thus maximizing potential QALYs for each clinical profile.

5.5. Sensitivity Analysis

We assess the robustness of the main findings to two key parameters: the latent SBP process variance σ^2 and the curvature of the stiffness attenuation function $\beta(w)$. These carry the most meaningful estimation uncertainty in the model: σ^2 is estimated from a single specialist cohort and directly governs how aggressively the optimal policy hedges against future SBP deterioration; $\beta(w)$ curvature is not directly observable from data. As we show below, however, the two parameters behave very differently. The σ^2 sensitivity materially affects the *magnitude* of the HTNMP advantage over BHS. The $\beta(w)$ curvature sensitivity — reported separately in Section 5.5.1 — turns out to be negligible: varying curvature across a 12-fold range (λ from 0.25 to 3.00) moves the QALY gain by at most 0.023 percentage points for males and 0.011 pp for females. The dominant source of uncertainty is therefore σ^2, analysed below.

Scenarios and results. Table 7 reports the three scenarios evaluated for each sex together with the corresponding QALY outcomes. The ABPM-implied upper bound is included as an external reference to confirm that even the high scenario remains far below any physiologically plausible upper limit: the ABPM-implied process variance of approximately 173 (female) and 161 (male) mmHg^2 is 30–41 times the Kalman MLE, because 20-minute ABPM readings capture circadian rhythms, postural shifts, and autonomic variability that average out entirely over a three-month epoch. The high sensitivity scenario at σ^2 = 8.60 / 5.91 mmHg^2 sits at roughly 5% of that bound, confirming the range is conservative.

Table 7 Sensitivity of QALY outcomes to the latent SBP process variance σ^2, by sex and scenario. Gain = (HTNMP − BHS) / BHS × 100%. BHS QALYs at baseline: female = 26.6061, male = 22.9109. The ABPM upper bound is included as an external reference only and is not used in the MDP.

		Risk-Free Female			Risk-Free Male		
Scenario	**Justification**	σ^2	HTNMP	Gain (%)	σ^2	HTNMP	Gain (%)
Low (−50%)	Conservative lower bound	2.87	26.9995	**1.498**	1.97	23.2180	**1.397**
Baseline (Kalman MLE)	MGH cohort, 3-month interval	5.73	26.8462	**0.902**	3.94	23.1682	**1.123**
High (+50%)	Conservative upper bound	8.60	26.7310	**0.585**	5.91	23.1271	**0.911**
ABPM upper bound (ref.)	20-min Var − $\hat{r}$; not used in MDP	172.8	—	—	160.9	—	—

Note. $\hat{r} = \text{MASD}^2/2 \approx 73$ mmHg^2 is subtracted from the ABPM total variance to isolate the short-term process component.

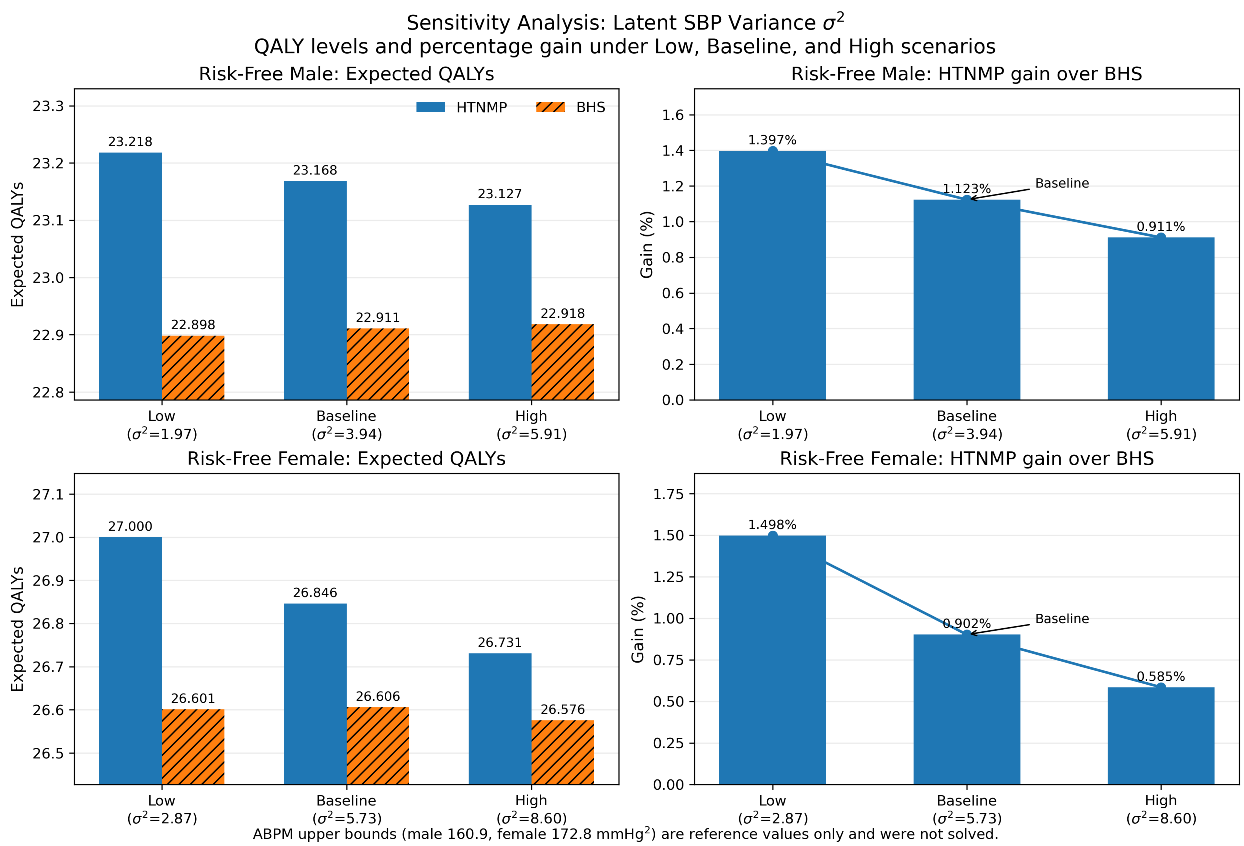


Figure 32 Sensitivity of QALY outcomes to the latent SBP process variance σ^2. Each row corresponds to one sex (Risk-Free Male, top; Risk-Free Female, bottom). *Left panels*: expected QALYs under HTNMP (dark) and BHS (light) for the Low, Baseline, and High σ^2 scenarios. *Right panels*: percentage QALY gain of HTNMP over BHS by scenario. Baseline Kalman MLE values are $\sigma^2_{\text{male}} = 3.94$ mmHg^2 and $\sigma^2_{\text{female}} = 5.73$ mmHg^2.

Three findings emerge consistently across both sexes.

Finding 1: absolute QALYs decline modestly with higher process noise, but the scale is small. For females, HTNMP QALYs fall from 26.9995 at the low scenario to 26.7310 at the high scenario — a drop of 0.268 QALYs, roughly 1% of the baseline value. For males the corresponding fall is 0.091 QALYs (23.2180 to 23.1271). Higher σ^2 raises within-period SBP volatility, which increases cardiovascular event risk under both policies; but because BHS QALYs fall by a similar absolute amount, the policies move together and the model's core structure is not materially altered.

Finding 2: the relative HTNMP advantage decreases as σ^2 increases. This is the most policy-relevant finding. For males, the QALY gain over BHS falls from 1.397% at the low scenario to 0.911% at the high scenario; for females it falls more sharply, from 1.498% to 0.585%. The mechanism is that under higher noise both policies are acting on noisier observations of the same latent SBP. BHS, which applies a hard threshold to the observed reading, becomes more trigger-happy — it prescribes more frequently because transient spikes push observed SBP above the threshold even when the latent level has not truly risen. This inadvertently aligns BHS more closely with HTNMP's intent, narrowing the gap. In other words, HTNMP's advantage comes from correctly distinguishing signal from noise, and that advantage is worth more when the signal is cleaner. The practical implication is direct: *the case for deploying HTNMP is strongest in structured specialist clinics with regular, consistent measurement conditions* — precisely the setting represented by the MGH cohort and by the baseline σ^2 estimates.

Finding 3: qualitative ordering is preserved across all scenarios. Despite the quantitative compression at high σ^2, HTNMP consistently outperforms BHS at every scenario and for both sexes. The baseline gain of 0.902% for females and 1.123% for males shrinks but does not vanish at the high scenario (0.585% and 0.911% respectively). The threshold structure documented in Section 4.1 is preserved throughout. The key numerical findings of the case study are therefore robust to sampling uncertainty in the Kalman estimates, even if the exact magnitude of the advantage depends on the noise level in the clinical environment.

Relationship to the covariance matrix. The sensitivity range spans the full off-diagonal implications of varying σ_h^2 in the quarterly covariance matrix

$$\Sigma_h(\Delta t) = \begin{pmatrix} \sigma_h^2 & \rho\,\sigma_h\,\sigma_{w,h,\mathrm{yr}} \\ \rho\,\sigma_h\,\sigma_{w,h,\mathrm{yr}} & \sigma_{w,h,\mathrm{yr}}^2 \end{pmatrix} \Delta t.$$

Varying σ_h^2 by ±50% changes the SBP–PWV covariance term $\rho\,\sigma_h\,\sigma_{w,h,\mathrm{yr}}$ by ±29% (since $\sigma_h \propto \sqrt{\sigma_h^2}$), providing an implicit joint sensitivity to the correlation structure as well. The PWV innovation variance $\sigma^2_{w,h,\mathrm{yr}}$ is held fixed at its ELSA-Brasil calibrated value throughout (Baldo et al. 2018). While PWV is a key state variable in the model, its innovation variance primarily affects the dispersion of long-run stiffness trajectories rather than the immediate treatment decision. Our robustness checks therefore focus on SBP process variance, which directly governs the sequential treatment trade-offs faced by the decision maker.

5.5.1. Sensitivity to the Stiffness Attenuation Function We assess robustness to the calibration of $\beta(w)$, the main channel through which PWV affects treatment effectiveness. Rather than changing its endpoints, we vary curvature while preserving $\beta(w_{\mathrm{ref}}) = 1$, $\beta(w_{\max}) = 0$, and monotonicity:

$$\beta_\lambda(w) = \left[\frac{w_{\max} - w}{w_{\max} - w_{\mathrm{ref}}(\mathrm{age})}\right]^\lambda, \qquad \lambda > 0,$$

where $\lambda = 1$ is the baseline, $\lambda < 1$ implies weaker attenuation (drugs remain more effective at high stiffness), and $\lambda > 1$ implies stronger attenuation. We evaluate $\lambda \in \{0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 2.00, 3.00\}$ and solve the full MDP for each value.

Table 8 reports the QALY gain of HTNMP over BHS for every tested scenario. The right-hand column provides a visual reference: each bar spans the full axis range used in the σ^2 sensitivity (0%–1.5%), so the reader can judge at a glance how little the gain moves relative to the variation seen under σ^2 perturbations.

Table 8 **HTNMP percentage QALY gain over BHS across all tested values of the $\beta(w)$ curvature parameter λ. The baseline functional form ($\lambda = 1$, shaded) is $\beta(w) = (w_{\max} - w)/(w_{\max} - w_{\text{ref}})$. Values $\lambda < 1$ give weaker attenuation (drugs more effective at high stiffness); $\lambda > 1$ give stronger attenuation. Gain = (HTNMP − BHS)/BHS × 100%.**

λ	Scenario	Male gain (%)	Female gain (%)	Visual (0% 1.5%)
0.25	Very weak	1.132	0.907	
0.50	Weak	1.128	0.906	
0.75	Mild	1.126	0.904	
1.00	**Baseline**	**1.123**	**0.902**	
1.25	Mod. strong	1.121	0.901	
1.50	Strong	1.119	0.900	
2.00	Very strong	1.115	0.898	
3.00	Extreme	1.109	0.896	
Range (max − min)		**0.023 pp**	**0.011 pp**	*vs. 0.912 pp (male) / 0.913 pp (female) under* σ^2

Notes. pp = percentage points. For reference, the σ^2 sensitivity (Table 7) produces a range of 0.486 pp (male: 1.397%–0.911%) and 0.913 pp (female: 1.498%–0.585%) — 20–80× larger than the $\beta(w)$ range. The gain column in the visual bar uses male QALY gain values.

The results are remarkably stable. The total range in QALY gain from $\lambda = 0.25$ to $\lambda = 3.00$ is only 0.023 percentage points for males (1.132% to 1.109%) and 0.011 percentage points for females (0.907% to 0.896%). Policy ordering does not change, and the dynamic-PWV model continues to outperform BHS across the entire range.

The mechanism behind this near-invariance is instructive. Increasing λ makes $\beta_\lambda(w)$ more concave: it attenuates drug effectiveness more strongly at intermediate PWV levels ($w \approx 8$–10 m/s) while leaving the endpoints unchanged ($\beta_\lambda(w_{\text{ref}}) = 1$, $\beta_\lambda(w_{\max}) = 0$). This shifts *both* the HTNMP and the BHS policy in the same direction — HTNMP adjusts its threshold accordingly, but BHS also prescribes more cautiously because the assumed drug effect is smaller. The two adjustments largely cancel, leaving the relative gain stable.

The incremental value of stiffness information therefore does *not* rest on the precise curvature of $\beta(w)$. The main insights — that dynamic PWV tracking dominates static modelling, that benefits concentrate in high-stiffness patients, and that the stiffness-aware policy prescribes fewer drugs to the stiffest patients — are structural features of the model that survive all tested λ values.

6. Implementation of Policies Under Full PWV Observability

Dynamic FI policy entails repeated optimization as a patient's state changes. However, since clinical practice requires that treatment protocols be simple, clear and easy for physicians to understand, dynamic optimization cannot be utilized.

Thus we show how Policies 1–3 are simple approximations of the Dynamic FI; all contain observed PWV information but restrict treatment sequencing, age dependence, or both.

6.1. Cross-Policy Comparisons

Table 9 Policy structures when PWV is observed.

Policy	Sequence	Age
Policy 1 (Age and Sequence Fixed FI)	⊠	⊠
Policy 2 (Age Fixed FI)	✓	⊠
Policy 3 (Sequence Fixed FI)	⊠	✓
Dynamic optimization (Dynamic FI)	✓	✓

Notes: Policy 1 has a strict order of medications and does not take age into account. Policy 2 has the ability for the provider to select from a flexible list of medications and has no regard for the patient's age. Policy 3 takes the patient's age into account but has a strict medication order. The dynamic optimization model allows for complete flexibility in the ordering of treatment and makes treatment decisions jointly based on medication choice, age, and arterial stiffness.

Table 10 State and decision variables as a function of sequence flexibility.

	Medication state	**Decision**
Sequence-Fixed	$\ell \in \{0, 1, \ldots, M\}$: number of medications currently taken	$u \in \{0, 1\}$: prescribe the next drug on the fixed list
Sequence-Flexible	$m = (m_1, \ldots, m_M)$, $m_i \in \{0, 1\}$: binary medication vector	$a \in \{0, 1, \ldots, M\}$: which remaining medication to prescribe

In relation to a representative risk-free female under the dynamic-PWV specification, Figure 33 depicts the relative medication prescription thresholds for: (i) the BHS guidelines, (ii) three of the applicable policy heuristics and (iii) the full-information, dynamic FI policy. Arterial stiffness is held at the specific age-based reference level, $w_{\text{ref}}(\text{age})$, so that for purposes of viewing how each policy structure maps between SBP and age-based treatment decisions; as measured by a comparison of the policy conclusions, it is evident that the simpler policy heuristics approximate the treatment logic of the full-information dynamic FI benchmarks, but do not provide identical treatment logic results.

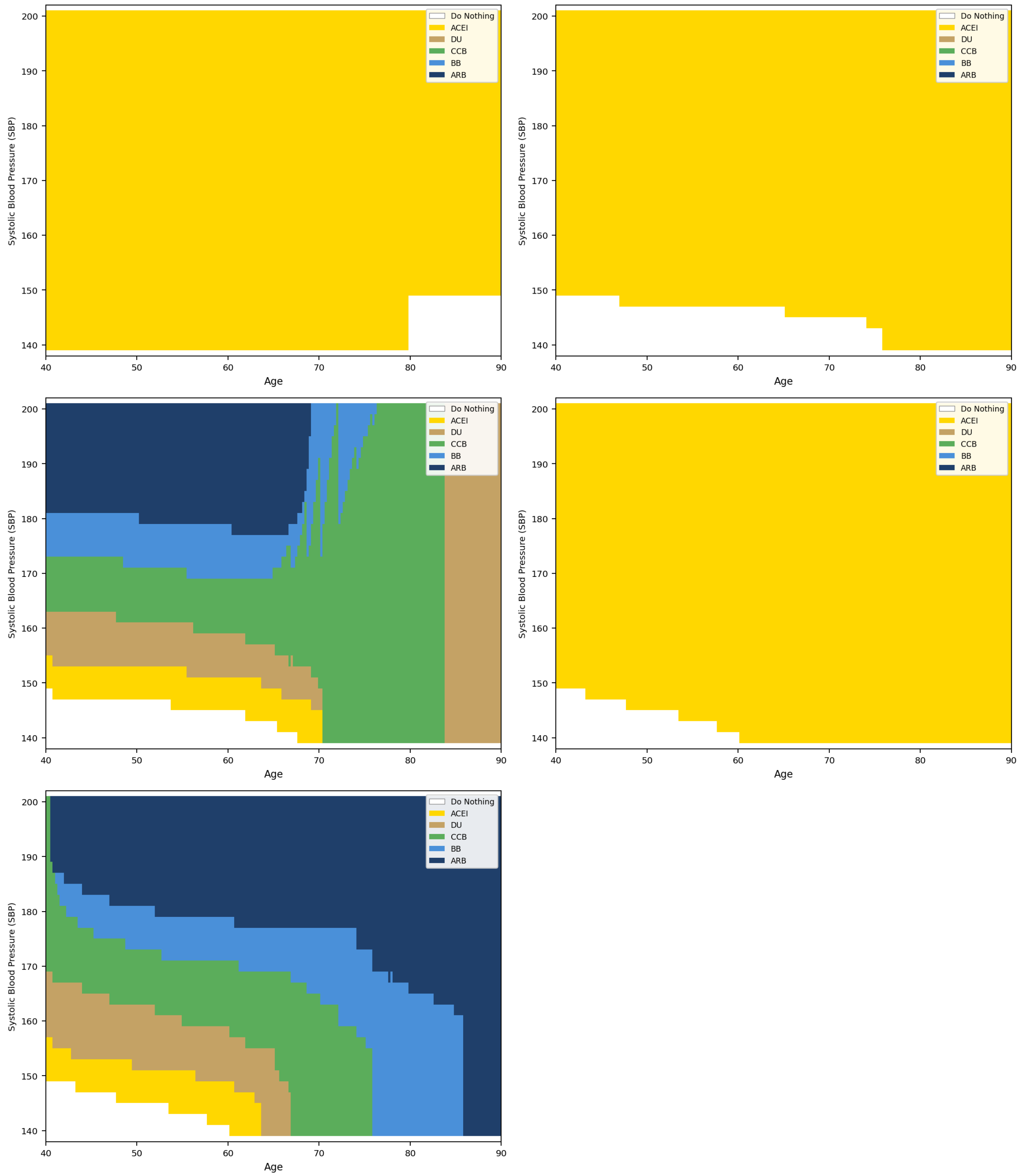


Figure 33 **Medication prescription thresholds by policy (risk-free female,** $\mathrm{PWV} = w_{\mathrm{ref}}(\textbf{age})$**). Panels (a)–(e): BHS guideline, Policy 1, Policy 2, Policy 3, and Dynamic FI.**

There is a distinct range of policy behavioral development over time as additional dimensions of decision making are applied, as seen in the five panels. More specifically, the figure delineates how specific aspects such as guideline rules, sequence flexibility, and age-dependent optimization contribute to determining treatment allocation.

BHS guideline. Panel (a) depicts the current guideline-based policy compliant with BHS, which dominates all of the treatment behaviour based on one class of drugs, with a small zone of no treatment recommended where low SBP is the criterion. Therefore, after crossing through the SBP threshold for treatment, the policy applies a single uniform threshold based on guidelines, which does not consider the patientś age when establishing its recommendations. This somewhat rigid implementation of guidelines demonstrates a preference for simplicity and clarity when developing these rules over optimising based upon the conditions at hand.

Policy 1 (sequence fixed, age frozen). Panel (b), shows the current optimal policy, fixed by medication sequence, keeping the patient aged 40 at the time of the decision. As a result, the policy has been established based upon the assumption that the patient will remain 40 years of age for the duration of his life while optimising; therefore, the resulting decision rule is stationary. For this reason, very minor changes to the treatment recommendation will occur along the age axis. Thus, as in the current guideline-based treatment policy, the preferred medication class for a patient collapsing most of the state space is one class of medications and does not permit any flexibility to modify treatment behaviour as the cardiovascular risk of the patient evolves throughout his or her lifetime.

Policy 2 (sequence flexible, age frozen). In Panel (c), we see a significant increase in the richness of the policy map when we allow flexibility in the choice of medication sequence for all ages, as previously described for 40 years old (state space). The expansion consists of an expansion that includes bands (a mixture) of different classes of medications on the SBP-age plane (the plane defined by SBP and Age) when we allow flexibility in the medication choice among classes of medications. In this case, the efficacy of medications are checked (based on SBP) for levels of SBP as it increases. For example, as the SBP level increases from a lower level to a higher level, the lower bands of medications become occupied by higher level medications. Each band will have different (better or worse) advantageous risk-to-benefit profiles for that medication for intermediate SBP ranges. Therefore, while the SBP-age system includes multiple bands through the addition of flexibility, there is no inclusion of age in the state space (there is no preferred band for medications available once the patient reaches that point in their life). As a result, as older patients continue to age and progress through this SBP-age system, the differences between bands in terms of best-to-worst locations become less important than the many bands that are present.

Policy 3 (age in state, sequence fixed). Panel (d) shows the differences in how the treatment threshold(s) for the SBP-age disease will change as patients become older when we add age back into the state space. By introducing age back into the state space, we can shift the treatments as a patient matures while keeping the treatment threshold steady. However, the patient must also be older than the established treatment threshold, and thus the overall boundaries of the treatment sector are consistent across all levels of the SBP-age sector; because of the linearity of the sequential treatment constraints, they will remain consistent. Even though the treatment constituents of the SBP-age sector all show that as a patient ages, their risk of cardiovascular disease increases, they cannot fully take advantage of this information, because they cannot use the order of their medication "¿" in making decisions for medications anymore.

Dynamic FI. In Panel (e), the age dimension has been added to the state space and there is total flexibility in the choice of the sequence of medications, resulting in the most adaptive and richest policies. Treatment regions have been identified for the various classes of medications with the boundary lines of these regions moving systematically down in age, illustrating the increased risk for coronary disease resulting from vascular ageing. Furthermore, greater-order medications are found at increasingly lower systolic blood pressure levels with increasing age, indicating that as a person ages and is at greater risk for coronary disease, he/she receives earlier and more aggressive interventions. Overall, this figure illustrates the structural relationship between the two

dimensions of variability. Sequence variability creates the decision space in which alternative medications can be selected, while age variability establishes the point in time when each medication is optimally prescribed. Neither dimension alone can reproduce the total complexity observed in a fully information dynamic policy (FD). It is only when both dimensions are incorporated that the model can produce a fully adaptive treatment map that is responsive to systolic blood pressure levels and the evolving risk of cardiovascular events for the patient.

Therefore, Policies 1 – 3 provide progressively richer implementable approximations to the full-information policy. Each policy has only restricted one of the two dimensions of variability, while continuing to utilize an individual's physiologic variables (e.g., PWV) to isolate the practical trade-offs between decision complexity and clinical performance.

In the following section, we will evaluate the clinical efficacy of these implementable approximations by comparison of QALYs.

6.2. Overall QALY Comparisons

Having characterised the structural differences between the policy designs, we now evaluate their clinical implications. Specifically, we compare the expected discounted quality-adjusted life years (QALYs) generated by each policy relative to the BHS guideline.

This comparison serves two purposes. First, it quantifies the performance improvement achievable by dynamic optimisation under full information. Second, it evaluates how closely the simpler, implementable policies (Policies 1–3) approximate the performance of the fully adaptive Dynamic FI rule.

Results are reported for four representative patient profiles: risk-free female, diabetic female, risk-free male, and smoking male.

Having characterised the structural properties of each policy, we now quantify their clinical value by comparing expected discounted quality-adjusted life years (QALYs) across the four policy structures, using the British Hypertension Society (BHS) guideline as the common baseline. Results are reported for four representative risk profiles — risk-free female, diabetic female, risk-free male, and smoking male.

Overall improvements relative to BHS. Figure 34 reports the percentage improvement in expected QALYs relative to the BHS guideline for each policy structure.

Table 11 Discounted lifetime QALYs under each policy structure

Patient Profile	BHS	Policy 1	Policy 2	Policy 3	Dynamic FI
Risk-Free Female	25.775	26.061	26.083	26.129	26.147
Diabetic Female	24.329	24.271	24.281	24.410	24.416
Risk-Free Male	22.548	22.735	22.769	22.765	22.796
Smoking Male	21.354	21.500	21.531	21.513	21.546

Notes: Values report expected discounted QALYs under each policy structure. Policies 1–3 correspond to guideline-style treatment rules with increasing structural flexibility. The dynamic optimisation model allows both flexible medication sequences and age-dependent decision rules. All results correspond to the dynamic-PWV specification.

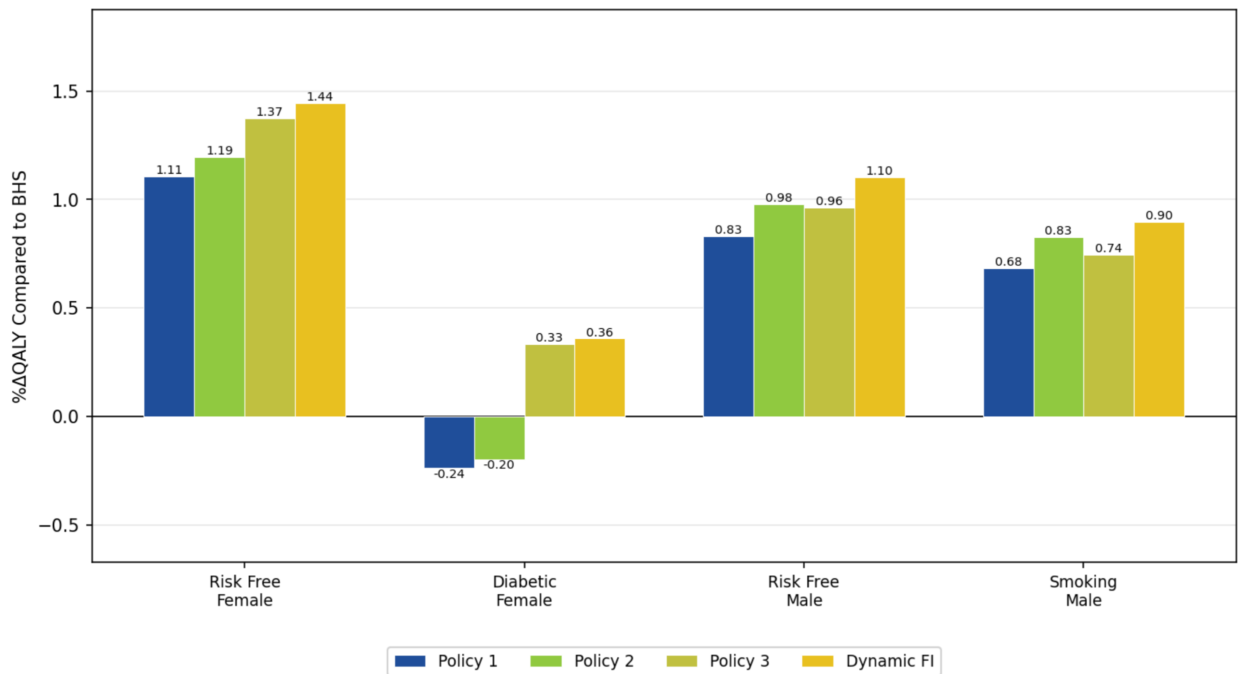


Figure 34 **Percentage improvement in expected QALYs relative to the BHS guideline under each of the four policy structures for four representative patient profiles. The right panel zooms in on Policies 1 and 2 to highlight their smaller gains.**

6.2.1. Policy Performance Relative to BHS Guidelines Figure 34 reports the percentage improvement in expected QALYs relative to the BHS guideline across the four policy structures. The policies progressively incorporate additional decision dimensions: Policy 1 optimises treatment thresholds with a fixed medication sequence and no age dynamics, Policy 2 introduces sequence flexibility, Policy 3 incorporates age-dependent risk, and the dynamic optimisation model jointly optimises both dimensions over time.

Across all profiles, the dynamic optimisation model consistently delivers the largest improvements in expected QALYs, followed by Policy 3, Policy 2, and Policy 1. This monotonic pattern indicates that each additional modelling dimension contributes incremental value.

For the risk-free female profile, improvements increase steadily from approximately 1.11% under Policy 1 to 1.19% under Policy 2, 1.31% under Policy 3, and 1.44% under the dynamic optimisation model. A similar ordering appears for the male profiles, although the magnitude of improvements is smaller. For the risk-free male profile, gains rise from roughly 0.83% under Policy 1 to 0.98% under Policy 2, 0.96% under Policy 3, and 1.10% under the dynamic model. For the smoking male profile, improvements increase from approximately 0.68% to 0.82%, 0.74%, and 0.90% respectively.

The diabetic female profile highlights the importance of modelling age-dependent cardiovascular risk. Policies 1 and 2, which ignore age dynamics, perform slightly worse than the BHS guideline (-0.24% and -0.20%). Introducing age into the decision state (Policy 3) reverses this pattern and produces a positive improvement of approximately 0.33%, while the dynamic optimisation model increases the gain slightly further to roughly 0.36%. This result suggests that policies calibrated under age-invariant dynamics may underperform guideline-based treatment when applied to high-risk patients whose cardiovascular risk evolves more rapidly over the life cycle.

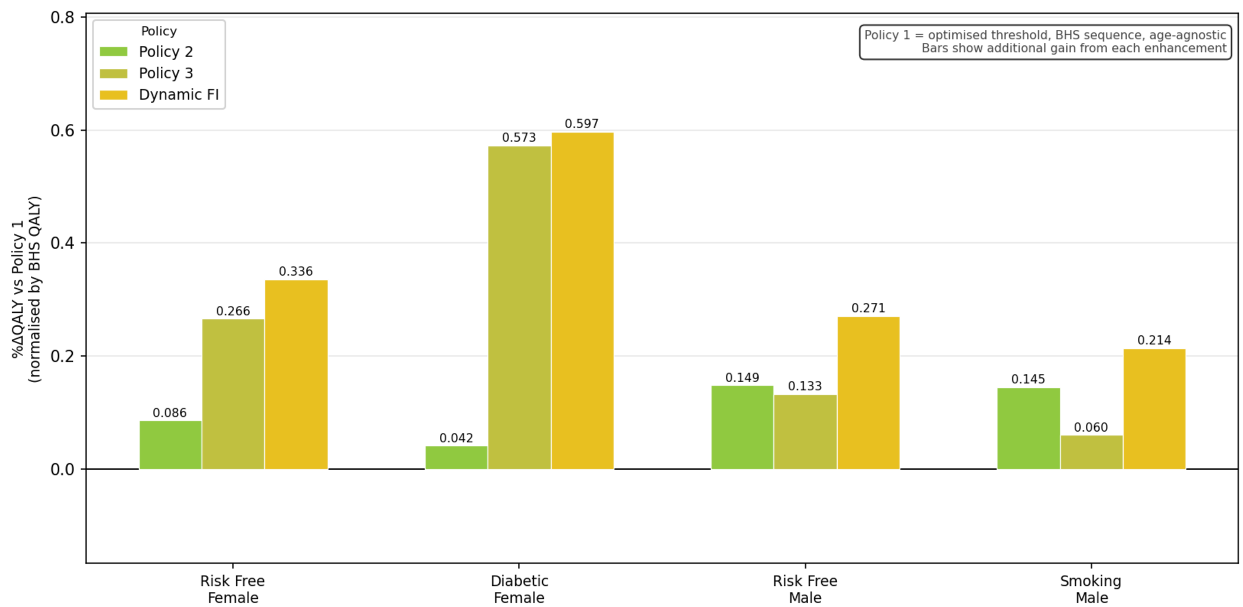


Figure 35 **Incremental QALY gains relative to Policy 1 from sequence optimisation (Policy 2), age personalisation (Policy 3), and their combined effect in the dynamic optimisation model.**

6.2.2. Decomposition of Policy Contributions To better understand the mechanisms behind the improvements observed in Figure 34, Figure 35 decomposes the QALY gains relative to Policy 1 into two components: the marginal contribution of sequence flexibility (Policy 2) and the marginal contribution of age personalisation (Policy 3). The dynamic optimisation model reflects the combined effect of both dimensions.

Across all profiles, both mechanisms contribute positively to performance improvements, although their relative importance varies substantially across risk groups.

For the risk-free female profile, both dimensions play a meaningful role: sequence flexibility contributes approximately 0.087%, while age personalisation contributes about 0.266%, yielding a combined improvement of roughly 0.336% under the dynamic model.

For the diabetic female profile, age personalisation is the dominant driver of performance improvements. Moving from Policy 1 to Policy 3 produces a gain of approximately 0.573%, whereas sequence flexibility alone contributes only about 0.041%. The dynamic optimisation model achieves a total gain of roughly 0.597%, indicating that most of the improvement arises from incorporating age-dependent cardiovascular risk.

For the male profiles, the contributions of the two mechanisms are more balanced. For the risk-free male profile, sequence flexibility adds approximately 0.149% while age personalisation contributes about 0.133%, producing a combined gain of roughly 0.271%. For the smoking male profile, the respective contributions are approximately 0.145% and 0.060%, with the dynamic model achieving a total improvement of about 0.214%.

These results suggest that age personalisation is particularly valuable for high-risk patient groups, whereas sequence flexibility plays a more prominent role when baseline cardiovascular risk is moderate.

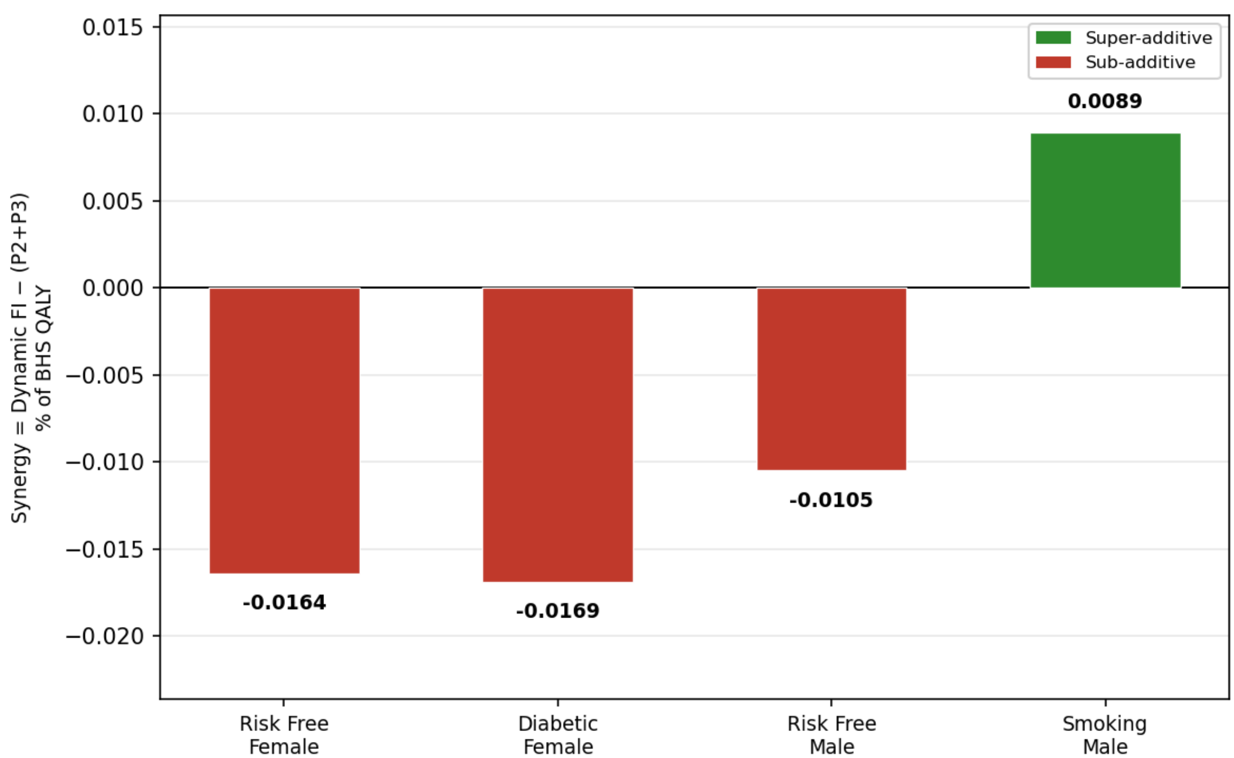


Figure 36 **Synergy between sequence optimisation and age personalisation in the dynamic optimisation model. Positive values indicate super-additivity, while negative values indicate sub-additivity.**

6.2.3. Interaction Between Sequence and Age Optimisation Finally, Figure 36 evaluates whether sequence optimisation and age personalisation interact when implemented simultaneously. Synergy is defined as

$$\text{Synergy} = \Delta_{\text{DynOpt}} - (\Delta_{\text{P2}} + \Delta_{\text{P3}})$$

where each Δ represents the improvement relative to Policy 1 as a percentage of the BHS benchmark.

The results indicate that interaction effects are small. Three profiles exhibit mild sub-additivity (-0.0164 for the risk-free female, -0.0169 for the diabetic female, and -0.0105 for the risk-free male), while the smoking male profile shows a small positive synergy of approximately 0.0089.

Because these interaction effects are very small relative to the total QALY gains, the two modelling extensions operate largely independently. Sequence flexibility determines which medication classes can be chosen, while age personalisation shifts the SBP thresholds at which treatment escalation becomes optimal over the life cycle. Consequently, the performance gains of the dynamic optimisation model arise primarily from the additive contributions of these two mechanisms rather than from strong interaction effects.

7. Conclusion

This paper proposes a new flexible decision-analytic approach to treatment for hypertension (high blood pressure) that directly incorporates arterial stiffness as an evolving physiological characteristic. By incorporating pulse wave velocity (PWV) into the Markov decision process, our model captures two important clinical mechanisms that are not currently included in existing hypertension decision models: 1) the impact of vascular aging on cardiovascular risk; 2) the effect of vascular aging on the effectiveness of antihypertensive medications. Our results indicate that taking arterial stiffness into consideration materially modifies the structure of optimal treatment policies and improves expected patient outcomes relative to guideline-based treatment. Specifically, the dynamically optimized treatment policies perform consistently better than those based on the British Hypertension Society (BHS) guidelines for all patient types; and the use of PWV information specific to the patient further increases the accuracy of the treatment recommendations and provides additional improvements in expected quality-adjusted life years (QALYs) for the patient. Particularly important is that PWV provides the largest benefit for patients who have a greater biological vascular age than their chronological age and for those patients who are close to treatment thresholds, where the clinical decision-making process is most sensitive to risk indicators. These results provide two complementary bases for the design of improved treatments: 1) dynamic optimization of sequences of decisions; and 2) accurate measurement of the underlying physiological state.

The findings demonstrate the ability to enhance the allocation of resources and targeting of treatment for patients with chronic diseases by integrating richer levels of information on an individual patient basis into decision support models. Furthermore, the results show that this framework provides a method to incorporate new medical laboratory technology (cardiovascular biomarkers) into the clinical decision support system to lead to more personalized and data-driven approaches to hypertension management.

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